# Corrections and Clarifications

FFGFT / T0 Time-Mass Duality

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## Preliminary remark

This document is the compact, running correction register of the FFGFT corpus. It supersedes the previous extended edition, which remains available unchanged as **Doc. 190-Archive** (`190_T0_Korrekturen_Archiv_De/En`) and contains, for every entry, the full justification, the faulty and corrected expressions verbatim, and all addenda. The table below lists *all* entries (C/K, P, R) with the affected documents and the subject in short form. The register remains append-only: new entries are appended here, and their elaboration is written directly in this document. The earlier rule not to revise source documents (R50) no longer applies since R133.

**Edition with correction notes (6 Oct. 2026).** Release v1.4.4 is the last edition of the corpus in which the documents carry the dated correction notes and correction boxes of the calculation review. It is preserved unchanged and records what was corrected when, and from which earlier value. Subsequently all documents are revised to make them easier to read: they will then contain only the corrected latest version, the text adapted to the corrections, without notes and correction boxes. The record of the corrections is this register and the v1.4.4 edition.

## .1 Register table

C/K = correction, P = clarification, R = revision/register entry. Full texts: Doc. 190-Archive.

| No. | Affects | Subject (short form; details: Archive) |
|---|---|---|
| C1 | 049 (HTML) | $\xi = \lambda_h^2 v^2/(16\pi^3 m_h^2)$ *[Addendum 28 July 2026, cf. R66: reason for the correction — $16\pi^3 = (4\pi)\cdot(4\pi^2) = S^2\cdot 2S^3$ is the sphere count of the compact $T^4$ boundary geometry (Doc. 310); the earlier version counted one 2-sphere too many]* |
| C2 | 116 (tab.) | $r_\tau = 25/9$ |
| C3 | 018 (Explorer) | $a_e = 2\pi^2/(f\cdot k)$ *(with $f = 1/\xi$, not a frequency; R147)* |
| C4 | 267 (DE+EN), 268 (DE) | **Factor 3 = three observable space dimensions:** the observer measures $k^2_{\text{obs}} = (n_1^2+n_2^2+n_3^2)/L^2$ (the fourth torus direction is time unfolding). *(Superseded by R139: the factor 3 is a setting, not a consequence of projection; cf. the note in Doc. 268 of 30 Sept. 2026.)* |
| C5 | 275 (script v3) | **All three fixed in v4** (11 June 2026): $T^4$ as kNN in genuine 4D coordinates with dimension-matched 4D nulls (result: $1{,}329 \pm 0{,}025$ *inside* the band $[0{,}690,\ 2{,}406]$); HLV ensemble with $\sigma = 0{,}05$ ($0{,}710 \pm 0{,}008$, in the band); the shell detect … |
| P1 | 116 / 158 | both correct; $0{,}001\,\%$ for external use *(superseded: $\Delta Q < 0.00003\,\%$ follows from no PDG masses; empirically $\lvert\Delta Q\rvert \approx 0.0003\,\%$, ladder masses $+0.16\,\%$; note in Doc. 116 of 30 Sept. 2026)* |
| P2 | several | $\hbar$ follows from $\xi$ via $L_0 = \xi\cdot\ell_P$ *(superseded by P34/P39: $\hbar$ is an SI conversion, not derived from $\xi$; cf. the note in Doc. 163 of 30 Sept. 2026)* |
| P3 | 173–176 | two parameters: $\xi_0$ (geom.) and $\xi_{\text{Higgs}}$ (alg.) *(superseded by R61: one parameter $\xi$; $\xi_{Higgs}$ withdrawn as a second parameter)* |
| P4 | 180–182 | $L_0$ = geom. fundamental length |
| P5 | several | mass structure from $\xi$; $\sim 1\,\%$ agreement *(refined by R138: $r_i$, $p_i$ are settings; the $\sim 1\,\%$ agreement is not a confirmation)* |
| P6 | 116 / 158 | Koide only with numerical values (non-closure, Doc. 193) |

| No. | Affects | Subject (short form; details: Archive) |
|---|---|---|
| P7 | 005, 008, 019, 086, 189, 202 | scale structure from the recursion $\mathscr{R}$ |
| P8 | 005, 008, 012, 014, 041, 060, 133, 141, 159, 181 | geometric scale adjustment / fractal correction |
| P9 | 005, 006, 042, 046 | columns “scaling class” + “SM correspondence”; quarks via $m = r\xi^p v$ |
| P10 | 009 | $K = R_{pe}/(4/3)^7$: an empirical anchor; factor $1{,}314$ defined implicitly, not derived |
| P11 | 025, 003, 133 | correct: $(1 - 275/4 \cdot \xi)$, $\Delta = 0{,}0002\,\%$; Doc. 003 and Doc. 133 use different models *(refined by R137 (i): unit-dependent agreement, coefficient $275/4$ not derived; open, R70)* |
| P12 | 022, 230 | two different observables: cumulative (Doc. 022) vs. elementary $\xi/(2\pi)$ (Doc. 230); $10^{-5}$ unreachable at any finite $N$ from Doc. 022 |
| P13 | 013, 025, 061 | $\xi_{\mathrm{FFGFT}}/\xi_{\mathrm{UIFT}} = m_e c^2/(\frac{16}{9}\ln 2 \cdot \mathrm{eV}) \approx 414\,684$; table in Doc. 013 (archived) reconstructed *(refined: with the UIFT definition, $\xi m_e c^2/(k_B T_{\mathrm{CMB}} \ln 2) \approx 418\,521$; $414\,684$ only with $k_B T_{\mathrm{CMB}} = (16/9)\xi$ eV; note in Doc. 013 of 30 Sept. 2026; unit-dependent, R137)* |
| P14a | 041 | fractal path lengthening (geometric path stretching); energy loss only an artifact when the path lengthening is omitted; path-length dependence remains (Doc. 026/149/182); the same path lengthening carries the CMB treatment in Doc. 268 (formerly tracked separately as P19) *(refined by R137 (ii): scale not stated, calibrated via $H_0$; the form $z \approx \xi \ln(d/\ell_P)$ in Docs. 181/182 is incompatible with $z > 1$, notes of 30 Sept. 2026)* |
| P15 | 061 / 041 | marked as a $\Lambda$CDM comparison quantity; the static universe has no cosmological $z$; CMB by stationary thermalization |
| P16 | 026 / 064 / 181 | $H_0$ is emergent (decompactified), not fundamental; implemented in Doc. 263, Docs. 026/064/181 noted as outdated *(Docs. 026/064/181 have carried notes since 30 Sept./1 Oct. 2026; R137 (ii)/(iii))* |
| P17 | 028 | the DE/DM relation is circular (gives 2.5 for any $\xi$); MOND partially derived ($\xi^{1/4}$ genuine, $K_M$ free); CMB relations unaffected; noted *(CMB and $\Lambda$ passages in Doc. 028 corrected or annotated on 30 Sept. 2026; classification R137 (i)/(iii))* |
| P18 | 267 | the two pictures are *not* distinguishable through the observed ratios; the decision is theoretical, not empirical (conceptual placement) |
| P20 | 267 / 268 | the *form* is fixed: a dimensionless $\xi$ ratio ($R_H/\ell_P$). |
| P29 | 268 | *Only partially answered.* The factor 3 from the 3D projection ($3 \cdot \{1, 6, 14, 26\}$) is plausible, but the sequence $\{1, 6, 14, 26\}$ is *read off the data* (Doc. 268, step 4), not produced forward. *(The factor 3 is a setting, not a consequence of projection; superseded by R139.)* |
| P30 | 268 | the *naive* spherical Bessel sum $C_\ell = \sum_n N_{\mathrm{BE}} \lvert j_\ell(k_{\mathrm{obs}} D_A)\rvert^2$ does *not* reproduce $\{3, 18, 42, 78\}$ – it washes out to a quasi-continuum dominated by low modes. |
| P31 | 268 | *Leading behaviour derived forward.* Forward, $\mathrm{T}^4$ gives the harmonic backbone $1 : 2 : 3 : 4 : 5$ (symmetric resonance $k^2 = 3n^2$). |
| P32 | corpus-wide | follows from P20: $H_0$ is an *imported* external calibration, not derived. *(Partly implemented: $H_0$ passages in the documents up to No. 300 and in the A series annotated, R137 (iii), R140.)* |

| No. | Affects | Subject (short form; details: Archive) |
|---|---|---|
| P33 | 009 / 042 / 181 | the *form* of $\xi$ is forward: the harmonic $4/3$ (perfect fourth, Doc. 042/159) gives the leading digit, $3$ (space dimensions) and $2^2{=}4$ (spacetime) are read off the geometry. |
| P34 | 181 / 013 | *3+1 is empirical input* – no theory derives the dimension count; the earlier "follows necessarily" claim (Doc. 181) is overstated. |
| P35 | corpus-wide (esp. 182, 250s, scale tables) | $\xi^N$ **triviality:** "$X = \xi^N$" is by itself *vacuous*, since any magnitude can be written as a $\xi$ power – an exponent alone is not a prediction (cf. P10, P20, P33: form forward, number input). *(On the addendum in the archive: $E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}} = 7.398$ MeV, R72; $\alpha = \xi E_0^2$ is a property of the anchor, not a prediction, R135.)* |
| P36 | 182 / cosmol. sector | $R_H$ **interpretation refined:** $R_H = c/H_0$ is the *Hubble length* – the local expansion rate expressed as a distance – *not* the size of the universe and *not* the observable horizon (the latter $\approx 3\,R_H$ in $\Lambda$CDM). |
| P37 | 009, 188, 271, 272, 275 | **Three levels clearly separated:** **(1) Microscopic – input:** $\xi = 4/30000$ (fractal dimension). |
| P38 | 275 | **Status clarification:** (a) $1/\varphi$ is a *pass-through point*, not a fixed point – the only fixed point of $r \mapsto r(1-\xi)$ is $0$. |
| P39 | 182, 267, 277, corpus-wide ($H_0/R_H$) | $H_0/R_H$ **status refined (shared reference point, not an FFGFT gap):** the relation Planck length$\leftrightarrow R_H$ (exponent $41/4$, $R_H/L_0 = 6{,}49 \times 10^{64}$) is derived from first principles by *no* framework – why the universe is $\sim 10^{60}$ Plan … *(Refined by R135/R141: the SI link runs through one anchor; $G$ follows in form and order of magnitude from $\xi$ and this anchor.)* |
| C6 | 004, 025, 030, 039, 041, 053, 061, 063, 068, 081 | **Redshift is achromatic.** The fractal path-lengthening is purely geometric and stretches *all* wavelengths by the same factor: $1+z = e^{\xi x}$, independent of $\lambda$. *(Refined by R137 (ii): achromaticity stands, the scale of the path lengthening is calibrated via $H_0$; $z = \xi x$ in Doc. 064 is dimensionally inconsistent.)* |
| P40 | 005, 006, 011, 012, 133, 275 | **Only ratios are exact.** Absolute values carry the fractal/SI correction, absorbed into the bridge constants $\nu$ (masses $m = r\xi^p\nu$), $E_0 = 7{,}398$ MeV ($\alpha$) and $3{,}521 \times 10^{-2}/2{,}843 \times 10^{-5}$ ($G$; `calc_De.py`); in ratios they cancel ($m_\mu/m_e = 206{,}768$ … *(As of 30 Sept. 2026: $G$ bridge constants $3.47222 \times 10^{-2}/2.88294 \times 10^{-5}$ in `calc_De.py`; refined by R135, R139, R141; the derivation of the $100$ in Doc. 133 does not hold, note of 30 Sept. 2026.)* |
| P41 | 267, 280, 221 (cosmological sector) | **Checked and not booked (degeneracy risk at the shared scale).** $\Lambda$CDM signal $\Delta v = c\,\dot{z}\,\Delta t/(1+z)$ over 25 years: $+6$ cm/s ($z$=0.5), sign change at $z_0 = 1.92$, $-14$ cm/s ($z$=4), $-20$ cm/s ($z$=5); ELT/ANDES order $\sigma \sim 2$ cm/s. |
| P42 | 282, 292 (lepton sector) | **One declared reference point: the ratio $m_\mu/m_e$.** The framework fundamentally keeps the purely theoretical derivation from $\xi$; the lepton sector receives – analogous to P39 ($H_0/R_H$ as shared empirical reference) – exactly *one* declared reference point … *(Comparison value $m_\tau$ in the archive text: $+0.43\sigma$ against PDG 2024, $1776.93 \pm 0.09$ MeV; cf. the note in Doc. 292 of 30 Sept. 2026.)* |
| P43 | 006, 011, 292 (`calc_De.py`) | **No rework; recorded as a note.** Verified: `calc_De.py` does *not* use $K_{\text{frak}} = 1 - 100\xi$ explicitly. *($E_0 = 7.398$ MeV $= \sqrt{m_e m_\mu / K_{\text{frak}}}$, not a separate route-2 value; the route-2 formula has been removed from Doc. 292, R135.)* |

| No. | Affects | Subject (short form; details: Archive) |
|---|---|---|
| R41 | 284 (cf. 270, 249) | **The loss is not the SI conversion.** Per Doc. 270 the Type I duality decompactification $S^1_m \to \mathbb{R}_t$ ($\tilde{T}\cdot m = 1$) is a *mode-preserving embedding* (lossless); compactness survives as the discrete spectrum and as physical observables ($T_{\text{rev}}$ … *(CHSH~ $\xi$ cannot be resolved with present hardware, R142/R143.)* |
| R42 | 270 (cf. 248) | **Not an open geometric question – it is the measurement.** In the compact view the four circles are symmetric (no space/-time distinction); time is not selected by the geometry but is *relational* – the symmetry-breaking *is* the measurement (the Type I decompact … |
| R43 | 285, 283 | **Corrected to containment.** $C_3 < A_5$ (3-fold and 5-fold both permute the same icosahedron, `recompute_reconcile.py`) and $T^7 = T^4 \times T^3$ ($7 = 4 + 3$), so HLV ⊃ FFGFT at the group/dimension level – the models are reconcilable, not inco … |
| R44 | 272, 190 (P35), 285 | **Sharpened (no contradiction).** $\varphi$ acquires meaning *not* inside FFGFT but as the derived eigenvalue of HLV's 5-fold / perp sector (golden inflation $\varphi^2 = \varphi + 1$) – a marker of the boundary HLV\FFGFT. |
| R45 | 285, 282–284 | **Sharpened.** A quasicrystal has two spectra: the *structure / diffraction* spectrum is pure point (sharp Bragg peaks, $\approx 21\times$ above a random null, `quasicrystal_information.py`) – that is the information and exactly HLV's substrate-distinguishabilit … |
| R46 | 296 (cf. 025, 026, 063, 156) | **Open check-point (not yet processed).** This intermediate-scale formulation potentially runs *counter* to the already established corpus line, which treats dark matter/energy not as an open signature but as a *derived consequence*: Doc. 025 reads dark matter as a … *(Reference line since restricted: $z \approx \xi \ln(d/\ell_P)$ is incompatible with the observed $z(d)$, note in Doc. 156 of 30 Sept. 2026; $H_0/\Lambda$ status according to R137, P39.)* |
| R47 | 285, 297, 298 (cf. 286, 284, 295) | **In the time sector: containment, not enlargement.** In unified notation – both $D6+$time, Marcel as $D6+$spiral-time, FFGFT in Hilbert space as $D6+$recursion-time – the spiral-time window is not additive but *internal*: a bounded section (window) of the unrolled re … |
| R48 | 304 (cf. 295, 301, 303) | **Note withdrawn, document made precise (not repaired).** The overstatement lay solely in the *mail* and was withdrawn without reservation; the document was written throughout from the FFGFT reading and already booked the bridge as restricted. |
| R49 | 304, 305 (cf. 295, 303, 272) | **Vocabulary cited, marked as analysis, abstract de-symbolised — notation not renamed.** A consolidated first-use footnote attributes U1/U2/U3 *and* $M_\tau/M_B$ to Krüger (2026) with the primary source (`doi:10.5281/zenodo.21302278`) and documented lineage (Me … |
| R50 | 025, 063, 061, 064, 078 (superseded by 306; cf. 003, 010, 295, 298, 302) | **Justification replaced, conclusion unchanged.** The conclusion (no singular beginning; a finite core with a minimal scale $L_0$ set by $\xi$) stands. *(The bound is $T_0 = \xi t_P$, not $d_1$; cf. the correction in Doc. 306 of 1 Oct. 2026, R90.)* |
| R51 | 049 (cf. 095, 306, 267, 302; notation) | **A notational relic, not a substantive contradiction — and nonetheless no longer authoritative.** *(a) The resolution:* in the original T0 view, $T_0$ never denoted the instant zero but the *smallest possible time interval*. *(Bound $T_0 = \xi t_P$ instead of $d_1$, cf. the correction in Doc. 306 of 1 Oct. 2026; according to the note of 30 Sept. 2026 the $a_\mu$ value in Doc. 049 is not supported.)* |

| No. | Affects | Subject (short form; details: Archive) |
|---|---|---|
| R52 | 101, 165 (numerical values; cf. 180, 182, 013, 250, 290) | **Values corrected; the main relation is unaffected.** *(i) Wrong $L_0$:* correct is $L_0 = \xi \ell_P = \frac{4}{30000} \cdot 1.616 \times 10^{-35}$ m = $\boxed{2.155 \times 10^{-39}\ \text{m}}$ — as in Doc. 180 (the dedicated derivation), Docs. 013, 181, 182, 187 … |
| R53 | 267 §518 (table row *source of the scale*; cf. 182, 165, P20/P35/P36) | **Corrected reading — and the symmetry becomes sharper, not weaker.** *(a)* On the FFGFT side, the table row *source of the scale* correctly reads not "from $\xi$" but **"set / calibrated to $H_0$"**. |
| R54 | 292, 306 (cf. 307) | **(a) $N_0$ is not a base unit.** $N_0 \approx 38.6$ is a *ladder-internal*, generation-counting factor ($N_g = gN_0$, Dok. 292); its value is *open* (P35: $100/e$, $100/\varphi^2$, $39$ do not hit it reliably). *(On (b), refined: the lower time scale is $T_0 = \xi t_P$, $d_n = 100\xi_n$ decreases monotonically from $d_1 = 1/75$; cf. the correction in Doc. 306 of 1 Oct. 2026, R90.)* |
| R55 | 292 §Cross-ladder coupling (cf. R54, P35, P40/P41) | **The law stands; its evidential breadth is one entry, not three.** *(i) The third entry is forced arithmetically.* From $m_\tau/m_e = (m_\tau/m_\mu)(m_\mu/m_e)$ a uniformly multiplicative correction satisfies $K^{p_3} = K^{p_1}K^{p_2}$, hence $p_3 = p_1 + p_2$ *identicall* … |
| R56 | 188 §Special role of $n = 3$ (cf. 189 §Derivation hierarchy, 248, 230/231/232, R55) | **The three elements are hereby booked uniformly as foundational assumptions.** *(i) The declared framework.* FFGFT posits three structural elements: the fundamental relation $\tilde{T} \cdot m = 1$ (Doc. 189, level 1), the topology $T^4$ (Doc. 248), and the order $Z_3$ … |
| R57 | A015 (cf. 020, 030, A-Series) | **Demand dissolved — $\xi$ is a fundamental parameter.** A proof is neither possible nor necessary; the demand was wrongly posed. |
| R58 | A142, A140 (cf. 127, 163) | **Gravitational quantisation not required.** $G = \xi^2/(4m_{\text{char}})$ is intrinsic to the torus geometry; a separate quantum gravity does not arise, because the geometry $T^4/Z_3$ is pre-ordered and not dynamical in the same sense as quantum-field-theoretic fields. *(Refined by R141: intrinsic applies to the form of $G$, not to the last digit of its numerical value.)* |
| R59 | A190 (Higgs sector; cf. A130, 011, 043, 101; P40) | *Subject: the $2.3\,\%$ deviation between $\xi_{\text{EFT}}$ from the one-loop Higgs matching and $\xi_0 = 4/30000$.* **Structurally explained [S].** Two causes: (i) SI values are scheme-dependent (1-loop, $\overline{\text{MS}}$), i.e. the EFT precision values are not theory-independent raw data; (ii) fractal corrections in $v$ are not cleanly separable from the loop structure of the SM … |
| R60 | A160, A165 (cf. 020, 022, 023, 230; P35) | **Both factors now geometrically derived [B].** (i) $2\pi$ is the circumference of the unit circle — the natural conversion between the torus winding phase and the measurement angle; $\xi/(2\pi)$ is thus the phase correction per measurement. |
| R61 | 174, 175 (cf. A142, 147, 162; R57, P35) | **Data-driven value selection — not a second parameter; withdrawn.** *(i) The genesis: $\xi_{\text{Higgs}}$* was selected *because* it lands in the measured window; the two-spaces doctrine is rationalisation after the selection, not independent justification. |
| R62 | 005 §Baryon worked example (proton) (cf. R50, P35) | **The anchor of the formula line is $\pi^2/2$, not $m_\mu$.** *(i) Diagnosis.* The missing factor identifies the error as a single symbol: $(\pi^2/2)/m_\mu = 46.705$ hits the missing factor to $0.07\,\%$. |

| No. | Affects | Subject (short form; details: Archive) |
|---|---|---|
| R63 | 253 §$\xi$ resonance heuristic (code: `rsa/factorization_benchmark_library.py`; cf. 024, 075, 076, 176; R57, P35) | **$\xi$ is inert in the "optimal" range; the optimisation switched the filter off.** *(i) Inertness.* The code threshold is $1/1000$. |
| P44 | 253, 190 (R63); cf. A135, P35 | **Pre-registered, tested, and refuted for this design; the window width is a property of the measurement setup.** *(i) Design.* Shor phase estimation, $12$ counting qubits (phase resolution $2^{-12} = 2.44 \times 10^{-4}$), $8$ shots, $10\,\%$ fully random outcomes, $300$ … |
| R64 | `2/python/t0_no_fallback_test.py`; B18 archive README; cf. R63, P44, P35 | **The procedure is right, the separating cut too coarse, and the label wrong.** *(i) What holds.* The counters `t0_success_count` and `fallback_success_count` are incremented at separate places (l. 376 resp. 280/300), `get_statistics()` report … |
| R65 | 024, 075, 076, 147, 176, 253 as well as the entire `rsa/` folder and the HTML tools under `2/html/`; cf. R61, R63, R64, P44, P35 | **No procedure distinguishable from Shor exists; the method claim is withdrawn.** *(i) What is withdrawn.* (a) The claim of an independent T0 resp. $\xi$ factorisation procedure. |
| R66 | 097 (DE/EN and the standalone versions under `2/texn/`), 073 (DE/EN); cf. 310, A142, A190, R59, R61, P35 | **The $16\pi^3$ count is right — but not for the reason given in Doc. 097.** *(The computation in Doc. 097 gives $1/(32\pi^3)$; cf. the note in Doc. 097 of 30 Sept. 2026.)* |
| C7 | the fourteen HTML pages under `rsa/` and `2/html/` (cf. R63–R65) | **Rebuilt, not annotated: the pages now say what they do.** No correction box was inserted; what was corrected is the computation, the labelling and the method text. |
| R67 | A040 (fractal correction), A270 (bulk exponent 36), Doc. 133 (older-corpus counterpart); cf. P35, R61 | *(i)* With $D_f^{\text{eff}} = 2.973$ (three decimals) the rounding interval leaves $n \in [98.3,\ 102.0]$ open; even $D = 2.973$ exactly gives $n = 100.12$. *(Value $1 - 100\xi$ confirmed by R135; form of $K_{\text{frak}}$ open again, R139; Doc. 133 carries a note of 30 Sept. 2026: the factor $100$ is set there, not derived.)* |
| R68 | 279 ($H_0$ chain), 308 ($\Lambda$ reading, cosmic sector); cf. P20, P39, R53, R61; Doc. 312 | *(i) Identification:* the $\Lambda$ function is carried by the derived quantity $\Lambda^* := 1/R_H^2 = (\pi/2)^2 \xi^{20}/\bar{\lambda}_e^2 = 5.218{\times}10^{-53}\ \text{m}^{-2}$ ("closure scale"). |
| R69 | 308 §$a_0$ derivation; cf. P20, R68; Docs. 306, 312 | *(i)* From $\partial \mathsf{T}^4 = \varnothing$ (Doc. 306) an isotropic torus yields a compact time cycle $\tau_c = 2\pi/H_0 = 91.9$ Gyr [K\|P20]; the energy quantum is exactly $\hbar H_0$, identical to the momentum quantum of the spatial cycles. |

| No. | Affects | Subject (short form; details: Archive) |
|---|---|---|
| R70 | 061 (temperature/CMB), A085 (natural units), A080; cf. Docs. 279, 310, R61 | $T_0$ scale gap |
| R71 | 309 (scale anchor, inheritance problem); cf. P20, P39, R69 | sharpening: side-taking in the Hubble tension |
| R72 | A010 (glossary), A080 (units), A265 (redshift), A130 (two $E_0$ routes); cf. A040, R67 | $E_0$: bare and corrected *(classification of $K_{\text{frak}}$ refined by R135; the third, $\alpha$-free route was withdrawn in A130 on 30 Sept. 2026)* |
| R73 | 026 (geometric cosmology), 279 (Casimir–CMB–$H_0$); cf. P20, P35, R68, R71 | $H_0$ notations, exponent $41/4$ |
| R74 | 025 (Cosmology), 063 (Cosmic), DE+EN each; cf. 267, 313 | lithium problem: mechanism, not solution |
| R75 | 024, 034, 075, 076, 147, 176 | $\xi$ name collision (search parameter now $\sigma$) and rewrite of the Shor documents (for Doc. 147 cf. R142, R143) |
| R76 | 318 | Scope of mass derivations |
| R77 | 041, 160, 318 | Classification of early coupling constant formulas *(numerical values corrected in Docs. 041/318: $\xi^{-1/3} = 19.57$; for $\Lambda_{\text{QCD}}$ cf. R122)* |
| R78 | 160, 005 | $\alpha_s$ and running coupling |
| R79 | 321, 319, 049 | Algebraic derivation of the $SU(3)_c$ gauge group |
| R80 | 321 | $SU(2)_L$, $U(1)_Y$ and $\sin^2\theta_W\vert_{\text{GUT}} = 3/8$ **[B]** |
| R81 | 323, 321, 160, 320 | Weinberg angle at $M_Z$ from $\xi$ |
| R82 | 322, 320 | Hilbert-space embedding and spectral theory *(cf. the notes in Doc. 322 of 1 Oct. 2026: $D_f = 3 - \xi$ does not follow from the 100-step recursion; self-adjointness R86/R145)* |
| R83 | 319, 321, 323, 322, 320 | Status update: closed bridges |
| R84 | 324, 321, 009 | Casimir derivation of $\xi$; $\xi$ not a G(6) spectral value *(physical Casimir part superseded by R136; spectrum of $V$ corrected in Doc. 324)* |
| R85 | 313, 325, 257, 321, 322 | Hawking mechanism closed microscopically |
| R86 | Docs. 180, 322, 314, 321, 325 | Self-adjointness of $\hat{F}$ closed on the $\mathbb{Z}_3$-invariant sector, full space open (R145); $\xi = \lambda_{\min}(\hat{F}_{D_4})$ **[B]** |
| R87 | Docs. 320, 322, 340 | $\Delta m^2_{32}$: numerical values corrected, dev. $+16.0\,\%$; candidate analysis added **[K]**. *Closed by Doc. 340 (R103): Galois derivation $\Delta m^2_{\text{atm}} = 120\, m_\nu^2$ from $GF(27)^*$ orbits, dev. $1.0\,\%$ **[K]**.* |
| R88 | Docs. 326, 329 | $n_{\text{thresh}}$ scale-dependent: $n(L) = L/(\sqrt{2}\ell_P)$; Matzke's scale $= 2\pi\ell_P/\ln 2$ **[B]**, universal number **[X]** |
| R89 | Docs. 180, 019, 201, 032 | $m_T = M_{\text{base}}/\xi$: one structure, sector-dependent base mass; range $= \xi \cdot \lambda_C(M)$ **[B]** *(value $5.220$ GeV in Doc. 032 only with the set $E_0 = 7500$ GeV, cf. the note in Doc. 032 of 30 Sept. 2026)* |
| R90 | Docs. 180, 019, 201 | Exponent in $L_0 = \xi^1 \ell_P$ derived from $m_T \sim 1/\xi$, not by analogy **[B]** |

| No. | Affects | Subject (short form; details: Archive) |
|---|---|---|
| R91 | Docs. 019, 201 | $\lambda$ in $m_T = \lambda/\xi$ is a base mass, not the Higgs quartic — label misleading **[K]** *(label corrected in the text of Doc. 019 on 30 Sept. 2026 and in Doc. 201 by addendum R91)* |
| R92 | Docs. 250, 190 | Doc. 250: $\xi/\ell$ is the coupling $g_T$, not $m_T$; Schwarzschild check for $L_0$ is an identity **[B]** |
| R93 | Docs. 325, 329 | $S_{\mathrm{BH}}(M_{\mathrm{coll}}) = 2\pi\,\mathrm{nat} = L_{\mathrm{Matzke}}/\ell_P$ in bits, $n_{\mathrm{thresh}} = S_{\mathrm{BH}}/\sqrt{2}$ (refined by R144); residual entropy of the endpoint **[B]** |
| R94 | Doc. 325 | $I_{\mathrm{sector}}/I_{\mathrm{thermal}} = \ln 3$ for all $M$ (refined by R144); structural constant of the $\mathbb{Z}_3$ geometry **[B]** |
| R95 | Docs. 330, 270, 314, 327 | Two wording refinements (flatness: $K{=}0$ pointwise via metric descent; s.a.: bounded *plus* symmetric, refined by R145: only on the $\mathbb{Z}_3$-invariant sector); Type-I theorem [sketch]→**[B]** (function pullback $p^*$); $D_4$ specificity test open **[S]** |
| R96 | Docs. 330, 332, 270 | Type-I follow-up refinement after second Krüger feedback: $\mathrm{Per}_{R_m}(\mathbb{R}_t)$ with per-period norm is isometrically isomorphic to $L^2(S^1_m)$, *not* to $L^2(\mathbb{R}_t)$ – pullback remains a change of representation, not a new sector **[B]**; Nash–Kuiper attribution in Doc. 330/R95(i) corrected (existence theorem for $C^1$, no $C^2$ non-embeddability claim) **[K]** |
| R97 | Dok. 324 | Vacuum operator correction following Doug Matzke's objection (IPI-Mail 26. Aug. 2026): Matzkes $V = N_1^+N_2^+N_3^+N_1^-N_2^-N_3^- = -64 \cdot P_{N1}P_{N2}P_{N3}$ is an involution in $\mathbb{Z}_3\mathbb{C}$ ($V^2 \equiv -V \pmod 3$) **[B]**; Dok. 324 incorrectly identified the Pp-product $P_{P1}P_{P2}P_{P3}$ (complementary sector, different ray: $\lvert\langle \mathrm{vac}_V \lvert \mathrm{vac}_{Pp}\rangle\rvert = 0$) with Matzke's $V$; numerically verified by `pruef_324_vakuum_involution.py` (22 Assertions); Trine-Theorem $T_k^3 = \mathbf{1}$ und $\xi$-Nicht-Spektralwert remain correct **[B]** |
| R98 | Doc. 333, 332, 315, 295, 285 | Two meanings of "unrolling" in the corpus distinguished and the recursion bound to the $\tilde{T} \cdot m$ duality (Doc. 333, 27 Aug. 2026): (i) Type-I (measurement act, lossy, prior corpus usage Doc. 285): $S^1_m \to \mathbb{R}_t$, winding number discarded; (ii) Type-III (canonical function pullback, lossless, Doc. 332/R96): $L^2(S^1_m) \cong \mathrm{Per}_{R_m}(\mathbb{R}_t)$, exact change of representation, $m$ and $t$ unroll together **[B]**. $K_{\mathrm{frak}}$ plays no role in the Type-III pullback; it sits in the choice of $R_m$ via the mass formula and becomes visible only at the SI transition **[B]**. The recursion $\xi_{n+1} = \xi_n(1 - 100\,\xi_n)$ (Doc. 295/146) follows necessarily from $\tilde{T} \cdot m = 1$: every mass change forces a time change, which feeds back recursively. Case B (stable particle, $K_{\mathrm{frak}} = 74/75$) is the projection of Case C (running recursion, log-spiral) onto the static approximation **[B]**. Doc. 332 reduced to a short cross-reference. |
| R99 | Doc. 334, 307, 306, 285, 174 | Superposition without time (Doc. 334, 27 Aug. 2026): (i) classical physics and QM inherit tacit instantaneity — time runs externally, calculations have no duration **[K]**; (ii) hidden inconsistency: state $\psi(t)$ is instantaneous, but probabilistic evaluation $\lvert\psi(t)\rvert^2$ requires real time for verification — this tension is absent from the formalism **[K]**; (iii) superposition is doubly timeless: no intrinsic "when", no verification without time expenditure; (iv) collapse has no duration in the QM formalism; (v) Pauli theorem does not apply to compact $\tilde{T}$ (angle variable); (vi) $\tilde{T} \cdot m = 1$ gives every mode an intrinsic time scale — superposition within one mode has scale $\tilde{T}_i$, cross-mode gives incompatible $\tilde{T}_i \neq \tilde{T}_j$ **[B]** |

| No. | Affects | Subject (short form; details: Archive) |
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| R100 | Doc. 336, 324, 329 | Notation updated to symmetric $\mathbb{Z}_3\mathbb{C}$ (after Matzke, IPI mail 28 Aug 2026): elements $a + bi$ with $a, b \in \{-1, 0, +1\}$ instead of $\{0, 1, 2\}$; $+1 + 1 = -1$ (ternary inline carry) **[E]**. Frobenius fixed points $= \{0, +1, -1\}$ = FFGFT sectors $\Delta w$ **[B]**; norm classes $\{+1, -1\}$ instead of $\{1, 2\}$; tripotent eigenvalues $\{0, +1, -1\}$ **[B]**. Verification scripts 324–327 updated to symmetric $\mathbb{Z}_3\mathbb{C}$ (all assertions passed). |
| R101 | Doc. 317, 336, 338 | $\xi = 4/30000$ derived as Galois number **[K]**: core identity $(r_\mu/r_e)^2/\xi = \lvert\mathrm{GF}(9)^*\rvert^2 \cdot 5^2 \cdot \lvert\mathrm{GF}(27)\rvert = 43200$ (pruef_331). All six winding numbers $(n_{\theta,g}, n_{\phi,g})$ from $\mathrm{GF}(3^n)$ orders; recursion $n_{\theta,g} = n_{\theta,g-1} + n_{\phi,g-1}$ **[K]**. |
| R102 | Doc. 338 | $1/\alpha = 3700/27 = 137.037$ (dev. 7.6 ppm) from Galois group orders **[K]** (pruef_332): new observation $m_e \cdot m_\mu = 54 = \lvert\mathrm{GF}(3)^*\rvert \cdot \lvert\mathrm{GF}(27)\rvert\,\mathrm{MeV}^2$ (0.016 %; Doc. 011 gives geometric derivation). $\lvert\mathrm{GF}(27)\rvert$ cancels; no $\xi$, no $v$, no $m_e$ as explicit input. *(Classification according to R135: the agreement is a property of the anchor $E_0^2 = 54$ MeV²; MeV is the SI bridge.)* |
| R103 | Doc. 339, 340 | Frobenius decomposition $GF(27)^* \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$ (Doc. 339): fixed points $\{+1, -1\}$ = massive sector (assignment massive sector vs. photon open **[S]**, cf. the note in Doc. 339 of 1 Oct. 2026); 8 three-orbits = gluons; $\mathrm{Orbit}_4 = 2^{-1} \cdot \mathrm{Orbit}_1$ **[B]**. Neutrino mass hierarchy (Doc. 340): $\Delta m^2_{\mathrm{atm}} = 120\, m_\nu^2$ (1.0 %) **[K]**; $\Delta m^2_{\mathrm{sol}} = (11/3)\, m_\nu^2$ (0.46 %) **[B]**; $p_{\mathrm{vent}} = 5/24$ via branching rule **[B]**; $\theta_{13}$: $(25\xi)^{1/3}$ (1.4 %) **[K]** and $2/3 \cdot R_\nu$ (7.4 %) **[K]**; $\gcd(13, 24) = 1$, spiral $\mathrm{lcm} = 312$ **[B]**. Check scripts: pruef_340 (13/13), pruef_340b (6/6). |
| R104 | Doc. 006, 338, 319, 340 | $v = m_e/((4/3)\xi^{3/2}) = 248.3$ GeV (0.85 %) **[K]**, computed with the Galois $m_e$ from Doc. 338 (with the measured $m_e$: $v = 248.9$ GeV, with the measured $m_e$ and $K_{\mathrm{frak}}$: 245.6 GeV; cf. R112 and the note in Doc. 372 of 1 Oct. 2026); $G_F = 1/(\sqrt{2}\, v^2)$ (1.7 %) **[K]**; $\tau_n \propto G_F^{-2} = 909$ s (3.4 %) **[K]**; $\Delta m = \frac{39}{8} m_e - (27/3700) \cdot \Lambda_{\mathrm{QCD}} \cdot \frac{5}{6} = 1.2912$ MeV (0.17 %) **[K]**: QCD part from $r_d, r_u, r_e$ (Doc. 006); EM part from $\alpha = 27/3700$ (R102), $\Lambda_{\mathrm{QCD}} = \hbar c/\mathrm{fm}$ (Doc. 319), $5 = \lvert A_5 : A_4\rvert$ (Doc. 340), $6 = r_u$ (Doc. 006). Check script: pruef_319 (7/7). |
| R105 | Doc. 006, 011, 070, 182, 231, 257, 285, 306, 307 | **Retrospective marker certification — documents created before the marker system (1 Sept 2026).** These documents predate the introduction of the marker system (A010) and carry no markers of their own. Their derivations are internally consistent and are necessary chain links for the Galois programme (Docs. 336, 338, 339) and further documents. Retrospective status: Doc. 006 (particle masses, winding numbers $r_i$, $p_i$) **[K]**; Doc. 011 (fine structure $\alpha$, $E_0$, fundamental relations) **[K]**; Doc. 231 (Hilbert space extension, $L^2$ structure) **[B]**; Doc. 257 (information unit, bit-energy scale) **[K]**; Doc. 285 (FFGFT-HLV dimension bridge) **[K]**; Doc. 306 (native time-energy reciprocity from $\tilde{T} \cdot m = 1$) **[K]**; Doc. 307 (time in state space) **[K]**; Doc. 182 (maximum universe scale from $\xi$, Schwarzschild radius and Hubble scale) **[K]**; Doc. 070 (mathematical structure: $K_{\mathrm{frak}}$ cancels exactly in ratios, absolute values carry the correction) **[B]**. Dependent documents may treat these results as a certified chain link by citing R105, without modifying the original documents. *(Restricted by R109, R135, R137, R138; the documents have been corrected in the text since R133.)* |

| No. | Affects | Subject (short form; details: Archive) |
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| R106 | Docs. 342, 343 | Primes forced by GF$(3^k)$ **[B]**: the physical primes $\{5, 7, 11, 13\}$ are completely forced by the prime factorisations of $3^k - 1$ ($k = 3, 4, 5, 6$); $p = 13$ is the worldwide unique prime with $\mathrm{ord}_p(3) = 3$. Euler factor threshold $p^* \approx 9.76$ ($s = 2$) = 7-limit; Galois product density at chance level from $p_{\max} \approx 30$ (pruef_342, pruef_344). |
| R107 | Docs. 343, 338, 162, 186, 328 | $\xi$ as resolution floor **[B/S]**: the limit $k \approx 3$–$4$ (Theorem F) holds for $\varepsilon = 1.05\,\%$, one fractal level ($\approx 100\xi$, Doc. 162). At $\varepsilon = \xi$ (torus-mode linewidth, Doc. 328) the Galois grid stays resolved to $k \geq 8$, $p^* \approx 87$. Coupling: Farey $Q_c = \pi/\sqrt{6\varepsilon} \approx 12.5$; Arnold $K^q$ vs. Farey $1/q^2$ — same tripartition, different law **[B]**; FFGFT with $K_{\text{eff}} = 2\pi\xi$ dynamically deeply subcritical, discreteness topological **[B]**. TFLN $Q > 10^6$ resolves $\xi$, not $\xi^2$. Doc. 186 preliminary note: locking **[B]**, Fibonacci **[B]**, B meson **[X]** corrected; renamed to `_De`/`_En`. Resonance factoriser capped at $N_{\max} \approx 1/\varepsilon$ (13–26 bit); quantum computers/photonics as analogue computers with QM precision, curve at $N = 21$ since 2012 **[X]** (pruef_343c–e). |
| R108 | Docs. 340, 342, 343 | Dirac/Majorana partition of the GF$(27)^*$ classes **[B]**: sector pairing $k \mapsto -k$ and Majorana criterion $k \mapsto k^{-1}$ (mod 13) partition the four primitive classes identically — $\{f_1, f_2\}$ (orbits $O_1, O_3$, quadratic residues mod 13) Majorana, $\{f_3, f_4\}$ ($O_2, O_4$) Dirac. Canonical: Majorana layer = kernel of the quadratic character mod 13. Numbering unified per the polynomial table of Doc. 342 (sector pairing $f_1 \leftrightarrow f_2$, $f_3 \leftrightarrow f_4$; negation $f_i \leftrightarrow g_i$); exponent list in Doc. 342 corrected ($f_3, f_4$ were the same orbit). Consequence for Doc. 340: $\nu_1 \in O_1$, $\nu_2 \in O_3$ Majorana; $\nu_3 \in O_4$ **Dirac neutrino [B]** — no Majorana phase, $m_{ee}$ from $\nu_1, \nu_2$ only ($\leq 14.4$ meV), inverted hierarchy more strongly excluded; sterile neutrinos drop out as a separate prediction. Doc. 340 revised directly (pruef_343f). Precursor: Doc. 007 derives $m_{\nu_1} = 4.54$ meV from $\xi^2/2 \times m_e$ **[K]** and motivates the double $\xi$ suppression physically (neutrino = almost photon + geometric decoupling); flavour hierarchy first in Doc. 340. Addendum inserted in Doc. 007. |
| R110 | Docs. 320, 022 | **Addenda** on $\nu_3$ Dirac neutrino (Doc. 343 Theorem D‴, R108): Doc. 320 (spectral theory) — fixed-point derivation $F_2, F_5, F_7$ provides no Majorana/Dirac character; this follows from the Galois algebra of Doc. 340/343. Doc. 022 (QFT-ML addendum) — $\xi^3$ sterile suppression in DUNE row not to be interpreted as a sterile signal; Majorana layer fully occupied by $\nu_1, \nu_2$. Authoritative: Doc. 340/343. |

| No. | Affects | Subject (short form; details: Archive) |
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| R111 | Doc. 344, 345, 346, 347, 348 | **Charge quantisation, Gell-Mann–Nishijima, and generation structure from GF$(27)^*$ (4 Sept. 2026).** Doc. 344: Su[0]=Vss[0] exactly — neutrino = vacuum of Clifford-Fock space **[B]**; sector pairing $k \mapsto -k$ has no fixed point among primitive classes ⇒ all neutrinos Dirac **[B/S]**; nEXO (5–20 meV) decides. Doc. 345: literature comparison Furey/Dixon/Froggatt-Nielsen; FN: $\varepsilon^{q_i}$ free vs. $\xi^{p_i}$ forced **[K]**; after Docs. 346/347 equal to Furey/Dixon at charge content. Doc. 346: two Galois properties classify all SM fermions **[B]**: (i) quadratic residue mod 13 ⇔ electrically neutral; (ii) $N$ mod $3$ ⇔ colour charge; forbidden combination QR+triplet algebraically forced; Legendre symbol $(k/13)$ gives $Q \in \{0, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1\}$ **[B]**. Doc. 347: $Q = I_3 + Y/2$ algebraically derived **[B]** (refined by R146: hypercharge of the charged leptons open): $Y = -1 + \frac{4}{3}$NQR from Legendre symbol (Doc. 346), $I_3 = \pm\frac{1}{2}$ from jE7 projector (Doc. 344); all four fermion charges exact. Theorem E **[S]** of Doc. 346 closed. Doc. 348: root orbits $f_1$–$f_4$ are genuine Frobenius orbits; 3 elements = 3 generations **[B]**. Trace bilinear form: $f_3 \times f_4$ (charged leptons × quarks) = permutation matrix (Gen.$2$ ↔Gen.$3$ forced) **[B]** (in Frobenius order $M^{(34)} = -I$, the swap is an ordering artefact; cf. the note in Doc. 348 of 1 Oct. 2026); $f_1 \times f_3$: entries $1/\sqrt{2}$ **[B]**, consistent with large PMNS angles. CKM complexity: $\delta_{CP} \approx 68^\circ$ not derivable from GF$(27)^*$ ($\mathbb{Z}_3$ phases give 120°); usable as phase anchor (same status as $v = 246$ GeV). Check scripts: `pruef_346`, `pruef_347`, `pruef_348`, all assertions passed. |
| R109 | Doc. 006 (restriction of R105) | **Neutrino part of Doc. 006 not tenable [X].** The "direct method" $E_i = 1/\xi_i$ was set up incorrectly in the worked example: with $f_i = r_i$, $1/(\xi r_i)$ gives $e:\mu = 1:0.42$ (inverse hierarchy, dimensionless); the MeV values shown alongside are measured values. The neutrino masses $9.1/1.9/18.0$ meV follow from three incompatible prescriptions ($\nu_e = m_e\xi^2$; $\nu_\mu = r_\mu\xi^3 v$; $\nu_\tau$ from none); the exponent $p_{\nu_\tau} = 25/9$ is $r_\tau$ copied into the exponent column. The Yukawa method $m_i = r_i\xi^{p_i}v$ remains valid for charged leptons and quarks ($< 1.2\,\%$). The **[K]** certification in R105 applies to Doc. 006 only for $r_i$, $p_i$ and the Yukawa method, not for neutrinos. **Clarification (30 Sept. 2026):** Both methods are exactly equivalent via $f_i = 1/(r_i\,\xi^{p_i+1}v)$ (transformation equation in Doc. 006) and give the same masses with fixed $r_i$, $p_i$; the only error was setting $f_i = r_i$ in the worked example. Doc. 006 has been corrected accordingly. Authoritative for neutrinos: Doc. 340 (R103, R108). Doc. 343 D‴ records the discrepancy **[S]**. |
| R112 | Doc. 006, 319, R104; Doc. 351 | $v = 246$ **GeV is not a measurement but an SM definition [Q] (5 Sept. 2026).** $v \equiv (\sqrt{2}\,G_F)^{-1/2}$ follows from $M_W = gv/2$ and $G_F/\sqrt{2} = g^2/(8M_W^2)$; only $\tau_\mu = 2.196\,980\,3(22)$ µs (MuLan) is measured, $G_F$ follows via $\tau_\mu^{-1} = G_F^2 m_\mu^5/(192\pi^3)\,(1+\Delta q)$ with $\Delta q = -0.44\,\%$ (QED $O(\alpha^3)$, hadr. VP with $4.7\sigma$ CMD-3 tension; Eberhart et al. 2026) and Sirlin absorption of EW corrections. The irrational factors $192\pi^3$ and $\sqrt{2}$ come from the SM formulation, not from measurement. Corpus-wide wording correction: "$v = 246$ GeV (measured)" → "$v = 246$ GeV (SM definition from $G_F$)"; the FFGFT value $v = 248.9$ GeV without or $245.6$ GeV with $K_{\text{frak}}$ (measured $m_e$; R104 with the Galois $m_e$: $248.3$ GeV) is a comparison of structure against convention. Convention-free test only at the $\tau_\mu$ level — **open [S]**. |

| No. | Affects | Subject (short form; details: Archive) |
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| R113 | Doc. 338, 317, 011, R100, R101; Doc. 351 | **Anchor $m_\mu/m_e$ is QED-dependent; comparison values are not theory-free [Q] (5 Sept. 2026).** $m_\mu/m_e = 206.768\,277(24)$ derives solely from muonium HFS (LAMPF 1999, Liu et al.); extraction via $\nu_F = \frac{16}{3}\alpha^2 c R_\infty (m_e/m_\mu)(m_r/m_e)^3$ — the anchor implicitly contains $\alpha^2$; Eides (2026): CODATA underestimates the theory uncertainty by a factor of 5–10 (271–515 Hz instead of 51 Hz). $\alpha$: Rb (Morel 2020) and Cs (Parker 2018) disagree at $> 5\sigma$; the CODATA value lies $0.4\sigma$ from the $g-2$ value, $1.2\sigma$ from Rb and $3.8\sigma$ from Cs (cf. the note in Doc. 351 of 1 Oct. 2026). $m_\tau$: two methods (threshold scan BESIII $1776.91$, pseudomass Belle II $1777.09$ with $m_{\nu_\tau} = 0$) differ by 0.18 MeV; PDG average $1776.93(9)$, FFGFT $1776.97$ lies $0.4\sigma$ — test undecided. Exact computation: the residual $43200 - 42753.12 = 446.9$ in $(m_\mu/m_e)^2$ is $240\,000\sigma$ of the CODATA uncertainty; no measurement error explains it; it is an empirical discrepancy **[K]** (Doc. 352, § 9–10); physical attribution (R113 bridge) open **[S]**. Wording correction: "model-independently measured" → "least model-dependent available value". Bookkeeping rule: every prediction states anchor, number of steps, corrections applied, extraction method of the comparison value. All 21 sources verified (Doc. 351) Three Galois identities compared in rational arithmetic (verification script `pruef_351_rationale_vergleiche.py`): (1) $(m_\mu/m_e)^2 = 43200$: residual $-1.034\,\%$ ($240\,000\,\sigma$), empirical discrepancy **[K]**, attribution open **[S]**. (2) $m_\mu/m_e$ vs. $\sqrt{43200} = 120\sqrt{3}$: residual $-0.52\,\%$ is *not an independent finding* — $\sqrt{43200}$ is irrational; the residual is exactly half the quadratic residual (artefact of taking the square root). (3) $m_e \cdot m_\mu = 54$ MeV$^2$ ($= |GF(3)^*| \cdot |GF(27)| = 2 \cdot 27$): residual $-0.016\,\%$ ($7386\,\sigma$), physical, 65 times smaller than the quadratic residual; no computational artefact (rounding uncertainty 2.2% of measurement uncertainty), not covered by measurement tolerance **[K]**. **Bookkeeping rule:** Galois identities should be compared only in their natural algebraic form — where no irrational numbers appear. |

| No. | Affects | Subject (short form; details: Archive) |
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| R114 | Doc. 351 (update), Doc. 352 (new); R113 | **Lepton residuals: lepton ladder, Galois and Koide compared; R113 attribution clarified [Q] (5 September 2026).** Doc. 352 compares three FFGFT structures (lepton ladder Docs. 006/317, Galois identities Doc. 351, Koide relation) with PDG and expresses all residuals in units of $\xi$. Main findings: (1) Koide matches PDG to $0.046 \cdot \xi$ — most precise prediction **[K]**. (2) Galois difference residual $-77.6 \cdot \xi = -2 \times (+39.1 \cdot \xi)$ lepton ladder algebraically forced **[B]**. (3) Generation-dependent corrections $\varepsilon_i$ (order $73$–$192 \cdot \xi$, decreasing with generation) explain transition $Q_{\text{bare}}(+8 \cdot \xi) \rightarrow Q_{\text{PDG}}(-0.05 \cdot \xi)$ to 101 % **[K]**. (4) $Z_3$ circulant: angle $\theta = 2/9$ from Doc. A110 **[K]**; amplitude $d/c = \sqrt{2}$ (Koide condition) and $\theta$ are orthogonal parameters **[B]** — $Q = \frac{1}{3} + \frac{d^2}{6c^2}$ does not contain $\theta$. (5) Status separation of the $\varepsilon_i$ (clarified 6 September 2026): $\varepsilon_i = m_i^{\text{PDG}}/m_i^{\text{bare}} - 1$ is a *computed residual* **[K]** from PDG **[Q]** against FFGFT bare **[K]** — not a number taken from QED literature; its $\xi$ scaling is observed **[K]**; the assignment to QED extraction + SI projection is an *attribution* **[S]**, not a citation; the reason for the falling structure $\varepsilon_e > \varepsilon_\mu > \varepsilon_\tau$ is open **[S]**. The QED self-energy formula $\delta m/m \approx \frac{3\alpha}{2\pi} \ln(m/\mu)$ (equivalently $\frac{3\alpha}{4\pi} \ln(m^2/\mu^2)$) in Doc. 352 § 10 is **[Q]** (standard formula) but yields only the order of magnitude, not the $\varepsilon_i$ values themselves. Column "Origin" in Doc. 352 § 9 → "Presumed origin [S]". Doc. 351 updated: three occurrences of "physical (R113)" → "empirical discrepancy [K] (Doc. 352, § 9–10); physical attribution (R113 bridge) open [S]". Verification script `pruef_352_leptonreste.py` (32/32) **[K]**. |
| R120 | Docs. 180, 201, 321, 307 | **Status of Maxwell dynamics: imported input, no derivation (14 September 2026).** An external reading attributed to the corpus a "sketched derivation of the full dynamical Maxwell system" with status **[S]**, citing Docs. 201 and 307. Check against the chapter files: neither document treats Maxwell. Doc. 201 takes the field-strength term $-\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$ over as a component of the extended Lagrangian; Doc. 180 explicitly sets the Standard Model Lagrangian including this term as its starting point; Doc. 307 contains only the Kaluza–Klein reading of charge quantisation. Retroactive status assignment: the kinetic term and the field equations that follow from it are **[E]** in Docs. 180 and 201 (established physics, imported) — neither [S] nor an open bridge. What the corpus does derive is the gauge group $U(1)_Y$ from the fourth torus direction **[B]** (Doc. 321, R80), charge quantisation and Gell-Mann–Nishijima **[B]** (Docs. 346/347, R111), and $\alpha$ from $\xi$ **[K]** (refined: Gell-Mann–Nishijima up to the hypercharge of the charged leptons, open according to R146; $\alpha$ via $\xi$ and the anchor $E_0$, R135). A derivation of Maxwell dynamics from the $T^4/Z_3$ geometry is neither claimed in the corpus nor listed as open. |

| No. | Affects | Subject (short form; details: Archive) |
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| R121 | Doc. 352 (§ 10), Doc. 369 (new) | **Origin of the $\varepsilon_i$: second reading without QED bridge (20 September 2026).** Doc. 352 § 10 takes the T0 ladder as fundamental and attributes the generation-dependent residuals $\varepsilon_i$ to QED extraction and SI projection (attribution **[S]**, R113 bridge open). Doc. 367 permits a second reading: the Koide formula with $\theta = 2/9$ is fundamental and parameter-free from the icosahedron; the T0 ladder $r_k\,\xi^{p_k}$ is then a rational approximation to the transcendental Koide amplitudes, and the $\varepsilon_i$ are the distance between cosine and power law — without PDG, without QED **[K]**. In this reading the $\varepsilon_i$ differ because the three cosine values differ; the decreasing trend needs no external physics. Both readings reproduce the same numbers. Status: the open point "why do $\varepsilon_i$ decrease" (R113/R114) is answered in Reading B; what remains open instead is why the rational approximation parameters are exactly the GF$(3^n)$ winding numbers **[S]**. The R113 bridge thereby moves from "needed" to "needed only in Reading A". Doc. 352 § 10 remains valid as Reading A; Doc. 369 § 4 places both side by side. *(Restricted: the exact pair $(\sqrt{2},\ 2/9)$ is excluded for $m_\mu/m_e$, $\approx 440\sigma$; cf. the notes in Doc. 369 of 1 Oct. 2026.)* |
| R122 | Doc. 041, 005 (Higgs mass, $\Lambda_{\mathrm{QCD}}$); Doc. 372 (§ 3.6) | **Higgs mass not derived (21 September 2026) [X].** The parameter table of Doc. 041 gives the Higgs mass as a power of $\xi$ times $v$ and lists the measured value beside it; recomputed, the formula yields barely a fifth of that. The same table gives $\Lambda_{\mathrm{QCD}}$ as another power of $\xi$ times $v$; the formula is off by almost two orders of magnitude. Doc. 005 gives the Higgs mass as top mass times golden ratio with a fractal correction factor; that factor would have to exceed the scale $\xi$ by several thousand, hence it is a fitted parameter, not a derivation. Both documents remain unchanged in the derivation (the computed values have been annotated there since 30 Sept. 2026, R133); the status of the Higgs mass in the corpus is **[X]**: no viable derivation. The $A_5$ matrix-element spectrum (Doc. 372) contains no matching value either. The remaining rows of the Doc. 041 table are unaffected; $v$ is governed by Doc. 006 (R104). **Status (Doc. 373, 21 September 2026):** For the Higgs mass there are now two compatible candidates **[K]**: the Galois chain linking $Z$, Higgs and top, and the trace rule, by which the Higgs particle is the residual stiffness of the lattice left over by the gauge bosons. Trace conservation, counting and normalisation factor are justified; together with the Weinberg angle of Doc. 372 the electroweak boson spectrum follows from $v$ alone at tree level. The formulas in Doc. 041/005 remain wrong. The trace rule applies to the lattice modes only; for the top the Galois chain applies, the reading of an equal fermion budget is discarded. Still open: $\Lambda_{\mathrm{QCD}}$ from $\xi$ and the choice of $v$ (R104/R112). |

| No. | Affects | Subject (short form; details: Archive) |
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| R123 | Doc. 160; Doc. 133, Clarification 11 | **Naming conflict $K_{\text{frak}}$ in Doc. 160 (21 September 2026).** Doc. 160 uses the label $K_{\text{frak}}$ for a hadronic factor that does not agree with the primary $K_{\text{frak}}$ of Doc. 133 (fixed in Clarification 11), lies on the other side of one, and derives from a Koide-type relation. It is a different quantity; in the sense of Doc. 160 it is to be read as a hadronic factor $K_{\text{had}}$, while the primary $K_{\text{frak}}$ remains that of Doc. 133. The factor was selected in Doc. 160 by a scan over combinations; the result for the lifetime ratio $\tau_\mu/\tau_\tau$ is therefore **[K]** with a selection caveat, not parameter-free. Doc. 160 remains unchanged (it has carried a note since 30 Sept. 2026, R140: $144/125 = r_\mu/r_\tau$ is the bridge factor from Doc. 158). Consequence: calculations using $K_{\text{frak}}$ (e.g. $\nu$ in R104 or a future derivation of $\tau_\mu$) use exclusively the value of Doc. 133. Open bridge: geometric derivation of the hadronic factor **[S]**. |
| R124 | Doc. 374 | **Withdrawal of the #740 statement (22 September 2026) [X].** The result section of Doc. 374 concludes that the bridge to Observer Patch Holography is closed under the reading $E_\star = E_P$. That statement goes too far: the coefficient has meanwhile been derived on the other side, while the length identity, the clock and a matching factor are still carried there as open, as are most of the dependencies of the item itself. What remains valid are the bridge equations B1–B4 and the absolute scale through the FFGFT chain; the reading is no longer posited on the other side but derived. Doc. 374 has been corrected directly; the theorem and the status statements are in Doc. 376. |
| R125 | Docs. 085, 176, 179, 186 | **Bibliographic data and one standard-physics statement in the photonics documents (24 September 2026).** Checking the cited photonics literature against the publisher records revealed four points that do not affect any derivation. (1) Doc. 085: "Photonics for sustainable AI" appeared in *Communications Physics*, not in *Nature*. (2) Doc. 176: the article number given for Wang et al. (2022) differs from the DOI record; the DOI is authoritative. The paper is a gedanken experiment with a simulation for a small example number; it supports the Fourier step with classical light conceptually, not the statements on speed and energy efficiency of the optical Fourier transform, which rest on the Optica source alone. (3) Doc. 186: the placeholder "K. Name et al." stands for Shimura et al. (TDK), as already given correctly in Doc. 187. (4) Doc. 179, SPDC section **[STANDARD]**: the perpendicular polarization of the two photons in type-II conversion follows from phase matching in the birefringent crystal, not from angular-momentum conservation. The documents remain unchanged; FFGFT statements are not affected. |

| No. | Affects | Subject (short form; details: Archive) |
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| R126 | Doc. 230 (sections on states and measurement) | **Cylinder representation of the qubit: placement in the literature and justification of the Born statistics (24 September 2026).** Doc. 230 introduced the cylindrical coordinates $(z, r, \theta)$ without classification and attributed the Born statistics of measurement to the unknown phase $\theta$. Refined: $(z, r, \theta)$ is the Bloch sphere in cylindrical coordinates; measurement drives the point deterministically to one of the poles, which pole is reached being decided by the hidden quantity $\lambda$ along the axis; $(1+z)/2$ is not due to $\theta$ but is a geometric quotient (axis fraction, Archimedes' area preservation, only for complex amplitudes), not at root a probability, consistent with Docs. 365, 367 and 370. The closest relative of the measurement reading is the hidden-measurement model. For multi-qubit systems the deterministic reading is retained but, by Bell and Kochen–Specker, is necessarily non-local or contextual; the $T^4$ reading satisfies this. Sources [Q] added to Doc. 230; measurement section, state comparison and numerical equivalence reworded accordingly (the earlier identification of $r^2$ with $\lvert\alpha\rvert^2$ was incorrect). |
| R127 | Doc. 043, 044; Doc. 261 (units table, $\varepsilon_0$ paragraph) | $\alpha = 1$ **as a reinterpretation, not as a choice of units (25 September 2026, refined 29 September 2026).** Doc. 044 (section "Why can $\alpha = 1$ be set?") and Doc. 043 justify $\alpha = 1$ by redefining the unit of charge; Doc. 261 adopts this as a convention for the written value. The result holds, the justification needs refining: $\alpha$ is dimensionless and does not change under a pure choice of units, because $e$ and $\varepsilon_0$ transform together. $\alpha = 1$ can, however, be written by reinterpretation: the step $e := \sqrt{4\pi\varepsilon_0\hbar c}$ moves the coupling into the normalisation of the field, so that it appears as one in the interaction term and as $1/(4\pi\alpha)$ in front of the field term; the physics stays the same. For FFGFT the unit independence of $\alpha$ is a prerequisite: $\alpha$ is derived as a dimensionless quantity from the dimensionless $\xi$ **[B]**. The physical invariant $\alpha$ obtained from $\xi$ is authoritative (Doc. 261). The documents remain unchanged (the justification section; convention errors have been annotated in Docs. 043, 044 and 261 since 30 Sept. 2026, R133/R140). |
| R128 | Doc. 221 (Signature 3, falsification criterion 3) | **No deviation of the supernova time dilation from $(1+z)$ at high $z$ (28 Sept. 2026) [X].** Doc. 221 predicts for $z > 2$ a light-curve stretching $e^{\xi_0 x} = 1 + z + z^2/2 + \ldots$ instead of $(1+z)$ and presents this as a discriminating test against standard cosmology. This does not hold: the series arises only from inserting the small-distance approximation $\xi_0 x \approx z$ into the exact law; moreover, the document uses two incompatible definitions of the redshift side by side. From $E \propto e^{-\xi_0 x}$ it follows exactly that $1+z = e^{\xi_0 x}$, and because of $\tilde{T}\cdot m = 1$ wavelength and clock rate are booked together, so the time dilation equals $(1+z)$ for all $z$ **[B]**, in agreement with Doc. 312 and A265. Light-curve stretching is therefore not a discriminator between FFGFT and expansion, only against tired light without time dilation. By Doc. 267 this holds for all frequency and clock-rate signatures of Doc. 221: they are degenerate and not falsification criteria; a static picture with $\tilde{T}\cdot m = 1$ is therefore not excluded. Doc. 221 remains unchanged. |

| No. | Affects | Subject (short form; details: Archive) |
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| R129 | Doc. 006, 046 (universal quantum number table) | **Unbooked early changes to the quantum number table (28 Sept. 2026, recorded retroactively).** The table with $(n, l, j, r_i, p_i)$ for all particles was changed during revisions in 2025 without a register entry: the prefactors $r_i$ for tau, charm and strange and $p$ for $\nu_\tau$ were replaced, and the boson rows (Higgs, $Z$, $W$, photon, gluon) were removed. The new $r_i$ are corrections; with the earlier values tau and charm lay clearly off the measured values. The $p$ value for $\nu_\tau$ is not tenable by R109; the boson masses have since been treated in Doc. 372 and 373. Clarification on reading the table: $(n, l, j)$ coincide per generation for leptons and quarks and fix the generation and the exponent $p_i$, but not the prefactor $r_i$. Whether the quark $r_i$ are geometrically forced remains open (Doc. 189, 373) [S]. Doc. 006 and 046 remain unchanged (they have carried notes since 30 Sept. 2026, R133, among them $p_\nu = p_\ell + 1$; on the classification of the $r_i$ cf. R138). |
| R130 | Doc. 343 (Theorem A) | **Theorem A: scope of the statement (29 Sept. 2026).** Theorem A derives $(1 - 3^{-s})\zeta(s) = L(s, \chi_0)$ in the one-dimensional analogue (decomposition by $n$ mod 3) and calls the result the spectral zeta function of the $T^4/\mathbb{Z}_3$ torus. For the orbifold of the corpus (fixed-point-free $\mathbb{Z}_3$ on $D_4$, Doc. 314/330) the spectral zeta function is one third of the four-dimensional Epstein zeta of $D_4$, with the Weyl pole at $s = 2$ and without the factor $(1 - 3^{-s})$ **[B]**. Theorem A remains valid as a statement about the 1D analogue or $L(s, \chi_0)$ **[B]**; the identification with the spectral zeta of the orbifold is withdrawn **[K]**. The same holds for Theorem B (Euler product by $\mathrm{ord}_p(3)$) and Theorem C (zeros from the factor $(1 - 3^{-s})$): valid as statements about $L(s, \chi_0)$ and arithmetic modulo 3, not about the spectrum of the orbifold. The question whether the GF$(3^k)$ reading has a counterpart on the four-dimensional object is decided negatively: the Euler factors of the orbifold have the same form for every odd prime, including $p = 3$ **[X]**. |
| R131 | Doc. 091 (appendix zeta regularisation) | **Zeta regularisation of the mode sum closed for the corpus lattice (29 Sept. 2026).** Doc. 091 carries the zeta regularisation of the zero-point energy only as a sketch, with a general $\zeta(-1)$ and no specified momentum lattice. For the momentum lattice $D_4$ of the corpus, $\zeta(-1)$ is expressed in closed form through values of the Riemann zeta function and is negative, so the Casimir term is attractive **[B]**. For the orbifold $T^4/\mathbb{Z}_3$ the value is one third of this, because the $\mathbb{Z}_3$ acts freely on all momenta $\neq 0$; the nine fixed points in position space are unaffected **[B]**. At equal volume of the fundamental cell, $D_4$ has the smallest magnitude among the comparison lattices, in the same order as the lattice energy in Doc. 380 **[K]**. The point carried as a sketch is thereby closed; the coefficient is lattice-dependent. Doc. 091 remains unchanged. |

| No. | Affects | Subject (short form; details: Archive) |
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| R132 | Doc. 010, 012, 013, 016, 041, 054, 063, 066, 086, 116, 189, 285, 337, 354; A145, A266 | **Calculations of the gravitational constant corrected (29 Sept. 2026).** Recalculating every place where $G$ is computed or inserted into derived quantities revealed calculation and formula errors in side calculations and conversions: missing $c^4$ or $c^2$ in the transition to SI, wrong conversion factors and exponents, dimensionally inconsistent formulas, wrongly derived intermediate values and overstated precision. The central formula $G_{\mathrm{SI}} = \xi^2/(4m_e) \cdot C_{\mathrm{conv}} \cdot K_{\mathrm{frak}}$ is not affected. The errors are corrected directly in the listed documents; each place carries a dated note with the previous value. In Doc. 066 the section on the factor between SI and natural units has been rewritten: $\xi(m)$ is a ratio of lengths and the same in both systems. *(Statement of the agreement with CODATA refined by R141.)* |
| R134 | Doc. 000, 001, 001a, 002 | **Acronym expansion and $\xi$ formula in introductory documents corrected (30 Sept. 2026).** In Doc. 000 (Introduction Part 3) the section on Volume 3 gave "Fractal Fine-Geometry Field Theory" as the expansion of FFGFT; the correct name is "Fundamental Fractal Geometric Field Theory" (corrected in De and En). In Doc. 001, 001a and 002 the keyresult box showed, alongside $\xi = \frac{4}{3}\times 10^{-4}$, the auxiliary formula $\xi = \frac{4\pi}{3} \cdot \frac{1}{4\pi\times 10^4}$; this evaluates to $\frac{1}{3} \times 10^{-4}$ (one quarter of the correct value) because the $\pi$ cancel to $1/1$ and the factor 4 in the denominator remains. The auxiliary formula has been removed from all six versions (De/En of 001, 001a and 002); as introductory documents they carry no note. |
| R133 | Doc. 003, 004, 005, 006, 007, 008, 010, 011, 012, 013, 014, 015, 016, 018 (Rev. 10–12), 019, 020, 021, 028, 032, 033, 035, 037, 040, 041, 044, 046, 049, 056, 059, 060, 061, 062, 063, 066, 068, 070, 077, 081, 086, 091, 101, 105, 116, 122, 124, 127, 132, 133, 134, 137, 143, 145, 146, 149, 150, 155, 158, 159, 162, 164, 167, 168, 169, 172, 175, 180, 182, 183, 188, 189, 202, 210, 230, 261, 281, 282, 285, 290, 292, 313, 315, 326, 338, 351, 352; A010, A040, A080, A130, A145, A220, A261, A265, A267, A283 | **Findings of the calculation review of older documents corrected in the text (30 Sept. 2026).** Recalculating older documents revealed arithmetic and formula errors, incorrectly copied formulas, wrong statements about the literature and facts, and predictions that are calculated correctly but excluded by measurements. Differences explained by older anchors or lower numerical precision are not counted as errors. The errors are corrected directly in the documents listed; each place carries a dated note with the previous value. Excluded predictions are not rewritten but annotated in place. The box at the start of each document lists the places. |

| No. | Affects | Subject (short form; details: Archive) |
|---|---|---|
| R135 | Doc. 011, 041, 044, 070, 133, 137, 210, 261, 263; A010, A130, A220 (cf. R72, R77, P40) | **Classification of $\alpha$, $E_0$ and $K_{\text{frak}}$ after the calculation review (1 Oct. 2026).** The agreement of $\alpha = \xi E_0^2$ with the measured value is a property of the anchor $E_0$, which provides the only link to the SI; it is no longer listed as an independent prediction, and statements such as "exact" are annotated in the documents. The earlier formula for an $\alpha$-free second route to $E_0$ has been removed (fitted exponent, dimensionally inconsistent) **[X]**; the two routes are the bare and the fractally corrected mean of $m_e$ and $m_\mu$. Their comparison with $\alpha$ fixes $K_{\text{frak}}$ and confirms the value $1 - 100\xi$ **[K]**; a small residual between the value required by $\alpha$ and the set value is not explained by measurement uncertainties and remains open. The comparison is not an $\alpha$-free measurement of $K_{\text{frak}}$. Closed $\alpha$ formulas from $\xi$ alone and further "second routes" to $E_0$ in older documents do not hold **[X]** (R77). |
| R136 | Doc. 009, 016, 025, 061, 063, 078, 091, 166, 263, 279; A260 (cf. R84) | **Casimir–CMB link: identity, not a measured confirmation (1 Oct. 2026).** Several documents used the Casimir pressure formula as an energy density; all have been corrected to the energy density. The ratio of Casimir to CMB energy density at $L_\xi$ is an identity, because $L_\xi$ itself is determined from the measured CMB energy density and $\xi$; the comparison value listed as "experimental" was the same formula with a rounded $L_\xi$. At $L_\xi$ the two densities do not coincide; equality lies at a larger plate separation. The link is a statement about scales **[S]**, not a confirmation. R84 (ii) stands for the algebraic Casimir operator; the sentence that the physical Casimir effect (Doc. 009) independently yields the same factor $4/3$ is superseded – there the factor is appended, not obtained from QFT **[X]**. Open: whether $L_\xi$ has a physical Casimir signature of its own. |
| R137 | Doc. 008, 025, 026, 028, 039, 063, 064, 143, 149, 243, 245, 246; A265 (cf. P11, P13, P16, P39, R70, K6) | **CMB temperature, redshift, $H_0$ and $\Lambda$: classification refined (1 Oct. 2026).** (i) The relations for $T_{\text{CMB}}$ from $\xi$ yield a pure number that matches the measured value only in the energy unit eV, and the coefficient in P11 has no geometric derivation; P11 and P13 count as a unit-dependent agreement, not as a derivation, and the question is open (R70). The relation $T(z)$ in Doc. 039 is excluded by measurements **[X]**. (ii) $1 + z = e^{\xi x}$ requires a length that is not stated and is fixed only through the measured $H_0$; $H_0 = \xi c$ (A265) and $z = \xi x$ (Doc. 064) are dimensionally inconsistent. K6 stands as a statement on achromaticity; the scale is calibrated (P39). (iii) Statements that $H_0$ lies within the measurement uncertainties (Doc. 026) or that $H_0$ and $\Lambda$ follow from $\xi$ alone (Doc. 245, 246) are not supported (P16, P39). The $\Lambda$ formulas in Doc. 028, 143 and 149 do not reproduce the observed value **[X]**. |

| No. | Affects | Subject (short form; details: Archive) |
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| R138 | Doc. 006, 009, 012, 028, 046, 070, 116, 149, 150, 186, 188, 189, 192, 193, 202, 210, 241; A150 (cf. P5, P6, P9, P10, P42, R105, R114) | **Precision tables of the mass ladder are not confirmations (1 Oct. 2026).** Older documents presented accuracy tables of the ladder $m_i = r_i\,\xi^{p_i}\,v$, up to "exact", as confirmation. Because $r_i$ and $p_i$ are fitted to the measured values or measured values were entered into the tables, these agreements are identities; the ratios follow from the ladder to a few digits. The derivation of the $r_i$ from an $(n, l, j)$ formula does not hold, so the $r_i$ remain settings (P6, P9); likewise the charm and top exponents, which the rule in A150 does not generate. Koide follows from the ladder only approximately (confirms R114). Not tenable **[X]**: the $\varphi$ mass ladder in Doc. 188 and the explanation of the $\alpha$ difference by the pentagonal shift of $f$ in Doc. 149/150; the structural formula for $v$ holds with unshifted $f$ (Doc. 383). The proton ratios in Doc. 009 and 028 are calibrated, not a confirmation (P10). Unchanged: the ladder as a structure with one declared reference point (P42) **[K]**. |
| R139 | Doc. 268, 291, 293 (cf. K4, P29, R67, R121) | **CMB factor 3 and the form of $K_{\text{frak}}$ open again (1 Oct. 2026).** (i) In Doc. 268 a projection onto three spatial directions cannot produce the factor 3 in front of the sequence $\{1, 6, 14, 26\}$, because the projected wave number never exceeds the full one. The factor is a setting; K4 and the part of P29 listed as plausible are superseded, and peak selection as a whole remains open. The input function does not contain $\xi$; the statement "input only $\xi$" is withdrawn. (ii) In Doc. 291 the statement that the correction candidate read off from the measured $\delta^*$ and $100\xi$ are indistinguishable rested on the median slope instead of the local slope; with the local slope they are distinguishable. The same tolerance has been carried over to Doc. 293, where the icosahedral leak is not compatible with $\delta^*$. The form of the fractal correction factor is therefore open again **[S]** (R67, R121); the value $1 - 100\xi$ remains confirmed according to R135. |

| No. | Affects | Subject (short form; details: Archive) |
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| R140 | Continuation of R133: Doc. 009, 022, 024, 025, 026, 030, 036, 039, 042, 043, 047, 051, 052, 053, 054, 055, 057, 064, 067, 074, 078, 080, 083, 089, 093, 095, 097, 114, 129, 131, 141, 144, 148, 152, 153, 154, 156, 160, 161, 163, 165, 166, 170, 181, 184, 185, 186, 187, 191, 192, 193, 201, 203, 204, 205, 207, 241, 243, 244, 245, 246, 247, 250, 251, 253, 257, 258, 259, 262, 263, 268, 274, 276, 279, 291, 293, 296, 297, 299; A070, A075, A110, A138, A142, A150, A155, A165, A180, A230, A250, A260, A263, A264, A266, A270, A271, A272, A273, A280, A284 | **Calculation review extended to the remaining documents up to No. 300 and the A series (1 Oct. 2026).** Procedure and marking as in R133: errors corrected in the text with a dated note, excluded predictions only annotated, literature references corrected without a note; in the A series there are only notes in the text, without a box. Casimir, $H_0$ and $\Lambda$ passages are included. QM documents and documents above No. 300 have not yet been reviewed (QM documents: R142; documents above No. 300 and A015–A050 reviewed on 1 Oct. 2026, notes in the documents; consequences also R141). The consequences for the classification are given in R135 to R139. |
| R141 | Doc. 012, 013, 016, 033, 063, 086, 116, 180, 285; A145, A220 (cf. R132, P40, R58; Doc. 383) | **Classification of the precision of $G$: residual of the one-anchor chain in $C_{\text{conv}}$ (1 Oct. 2026).** $C_{\text{conv}}$ is the SI conversion of the one-anchor chain from Doc. 383. The numerical value used in the corpus additionally contains the small residual of this chain, as Doc. 383 itself shows. The form of $G$ and its order of magnitude follow from $\xi$ and the single anchor **[K]**; the deviation for each choice of anchor is given in Doc. 383. The agreement with the CODATA value stated in the documents listed arises only with this residual and does not count as an independent precision confirmation; the documents carry a note to this effect. This refines R132 (stated agreement), P40 (bridge constants) and R58: "intrinsic from the torus geometry" applies to the form of $G$, not to the last digit of its numerical value. $K_{\text{frak}}$ in the $G$ formula remains unchanged. Open: a structural explanation of the residual of the chain **[S]**. |

| No. | Affects | Subject (short form; details: Archive) |
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| R142 | Continuation of R133 (QM documents): Doc. 020, 023, 023a, 023b, 034, 035, 073, 147, 162, 173, 174, 175, 178, 183, 230, 231, 232, 282 | **Calculation review of the QM documents up to No. 300 (1 Oct. 2026).** The errors are corrected in the text, each passage carries a dated note with the previous wording. Consequences for the classification: the fractal damping changes CHSH only at a level far below the measurement noise; it does not establish a local bound, the Bell violation remains (Doc. 023, 023a, 035), and the deviation from the Tsirelson bound cannot be tested with present hardware. The earlier confirmation by hardware measurements is withdrawn throughout Doc. 147 **[X]**. The lemma and the main theorem on the $\varphi$-QFT in Doc. 147 are not proven **[X]**; the equivalence remains a proof attempt **[S]**. The QND argument in Doc. 178 does not distinguish the interpretations. The gate-error prediction in Doc. 162 and several predictions in Doc. 035 (Higgs coupling, neutrino masses, mixing angle, hydrogen levels) are excluded or not tenable **[X]**. The systematic deviation of the $\pi$ gates (Doc. 034, 175, 230, 231; A070) is absorbed by the pulse calibration of real qubits and is therefore not directly measurable, not testable as stated before. There is thus currently no FFGFT deviation in the QM sector that can be tested with present hardware **[S]**. *(Made more precise by R157.)* |
| R143 | Doc. 022, 035, 148; A165 (cf. R142, Doc. 147) | **Unsupported CHSH data and the $\xi$ fitted to them removed (1 Oct. 2026).** Older documents relied on CHSH values from alleged hardware runs in 2025 that are not documented in any record, and on a value of $\xi$ fitted to them together with agreements built on it (neutrino, Rydberg and DUNE sections). These statements have been removed; each passage carries a dated note. Also removed is the statement in A165 that the CHSH deviation is measurable with present hardware. Authoritative is the only documented hardware measurement in Doc. 147: it confirms the Bell violation, and the $\xi$ effect cannot be resolved with it. Status: corpus value $\xi = 4/30000$ unchanged; no hardware-based confirmation of the $\xi$ effect **[S]**. |
| R144 | Doc. 325 (cf. R93, R94; Doc. 326) | **R93 and R94 refined (1 Oct. 2026).** R93 equated the Bekenstein–Hawking entropy at the collapse mass with Matzke's threshold. The two are related through the geometry of the collapse point but are not equal: the entropy equals Matzke's length in Planck units, the threshold is smaller by a factor $\sqrt{2}$, and this factor is the ratio of Schwarzschild radius to Planck length at that point. The identity remains exact **[B]**; only the identification is corrected. In R94 the ratio of sector to thermal information in equal units is $\ln 3$, not $\log_2 3$; mass independence remains **[B]**. Doc. 325 carries the notes and the correct register numbers. |
| R145 | Doc. 327, 322, 330 (cf. R82, R86) | **R86 refined: self-adjointness of $\hat{F}$ proved only on the $\mathbb{Z}_3$-invariant sector (1 Oct. 2026).** The $\mathbb{Z}_3$ pairing makes $\hat{F}$ normal; its sector values are pairwise complex conjugate, not real. $\hat{F}$ is therefore self-adjoint on the $\mathbb{Z}_3$-invariant sector **[B]**; on the full space only the real part $(\hat{F} + \hat{F}^\dagger)/2$ is self-adjoint, and using it is a choice **[S]**. The smallest eigenvalue $\xi$ of the $D_4$ sub-operator remains well-defined provided the $D_4$ sector lies in the invariant sector. The earlier counter-check did not distinguish paired from unpaired phases. Docs. 327 (Theorem 3, corollaries, verification script), 322 and 330 carry the notes. |

| No. | Affects | Subject (short form; details: Archive) |
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| R146 | Doc. 347, 346, 356, 357 (cf. R111) | **R111 refined: Gell-Mann–Nishijima not fully closed (1 Oct. 2026).** The hypercharge from the QR/NQR class alone does not separate charged leptons and quarks and gives the antineutrinos the wrong sign; the assignment must include the charge class modulo 3 and the particle/antiparticle distinction. The hypercharge of the charged leptons therefore does not follow from Theorem A of Doc. 347 and is open **[S]**. Charge quantisation (Doc. 346) remains **[B]**. The documents carry notes. |
| R147 | Docs. 018, 159, 186, 210 and others (cf. A135) | **Notation $f$: not a frequency; $f = 1/\xi$ cannot be set to 1 (2 Oct. 2026).** The symbol $f$ carries four meanings in the corpus: (i) the base winding number $f = 1/\xi$ (sub-Planck factor, anchor number) in the $g$-2 formulas, in $t_0 = t_P/f$ and in the formula for $v$; (ii) a frequency; (iii) the geometric structure factor $f(n,l,j)$ of the quantum numbers; (iv) further functions, among them the fractal correction function in $g = \eta(1+\xi f)$, $f(k_t)$, $f(\theta)$ and dependences that are not written out such as $f(\xi)$. Binding: where $f$ without an argument stands for a number, the base winding number is meant, unless the note in the document says otherwise (different in Docs. 033, 044 and in the $g$-2 formula of Doc. 159). In natural units $f$ is only another reading of the duality $T \cdot m = 1$: $f = t_P/t_0 = \Lambda_{\mathrm{TO}}/E_P$, and $\xi E_0^2 = 1$ gives $f = E_0^2$. As a dimensionless ratio, $f$ cannot be set to 1 with justification; by A135 only quantities that become identical through the duality ($c$, $\hbar$, $k_B$) or by normalisation ($\alpha$) can be set to 1. The places with an FFGFT meaning carry a note, and the numbered documents with a correction box also an entry there: Docs. 001, 001a, 002, 003, 005, 006, 007, 010, 011, 012, 013, 018 (parts 10–12), 019, 022, 023, 023a, 023b, 033, 035, 040, 041, 044, 046, 047, 051, 052, 054, 060, 062, 063, 070, 077, 081, 089, 101, 105, 114, 122, 132, 133, 137, 145, 147, 149, 150, 156, 158, 159, 160, 163, 168, 169, 181, 182, 186, 189, 190 (with archives), 205, 210, 231, 232, 270, 282, 285, 286, 287, 288, 291, 307, 316, 321, 382, 383, 384, 387; A series: A040, A070, A075, A138, A165, A266, A284. Standard mathematical uses (Shor function, structure constants $f^{abc}$, focal length, distribution and test functions, degrees of freedom) remain without a note. |

| No. | Affects | Subject (short form; details: Archive) |
|---|---|---|
| R148 | Continuation of R133 (addendum, mainly documents above No. 300): Docs. 267, 284, 295, 301, 304, 306, 308, 309, 310, 311, 317, 318, 319, 320, 323, 324, 328, 329, 335, 336, 339, 340, 341, 342, 343, 344, 345, 348, 349, 350, 353, 355, 358, 363, 364, 366, 368, 369, 371, 372, 373, 381; A020, A030, A120 | **Addendum to the calculation check (1 October 2026, recorded 7 October 2026).** R140 mentions the check of the documents above No. 300 without listing them; in addition, a few older documents whose notes are not in any collective entry. Procedure as in R133; the notes with the previous wording are contained in release v1.4.4. Consequences for the classification: The Higgs route does not lead to $\xi$; it remains a consistency check (Doc. 320, cf. R59); the $W/Z$ masses in Doc. 320 are inputs. GF$(9)$ is not a subfield of GF$(27)$; the exponent in Doc. 336 is a setting, the photon spin in Doc. 339 is not derived. Triality and orbifold sector are two different $\mathbb{Z}_3$ (Docs. 336, 341, 358). Theorem A in Doc. 349 and Theorem C in Doc. 363 are not proven; the derivation of the Koide amplitude (Doc. 353), the hypercharge of the Higgs doublet in Doc. 355 and the fermion statistics in Doc. 319 are open **[S]**. The quark prefactors in Doc. 373 are determined from measured values. In four dimensions fivefold symmetry is lattice-compatible (A020, A030). The Casimir effect contains no factor $4/3$ (Docs. 324, 371, A020). The galaxy tests in Doc. 308 are not passed but open or circular, respectively. |
| R149 | Docs. 006, 073, 382; HTML pages in `2/html/`, `rsa/`, `sig/` (cf. K7, R65, R148) | **Corrections in Docs. 006, 073 and 382 (8 October 2026).** (i) Doc. 382 computed the direct $g$-2 absolute formulas with a rounded $k_{\text{geom}}$; the values have been brought in line with those of Doc. 018 (Rev. 12). The ratio line and the $a_\tau$ prediction based on it are not affected. (ii) Doc. 006 compared the quark masses with older reference values; the comparison now uses PDG 2024, as in Doc. 384. The deviations become larger, for the up quark within the large uncertainty of this value; the quark prefactors remain determined from measured values **[S]** (R129). (iii) The numerical simulation parameter in Doc. 073 is called $\sigma$ instead of $\xi_{\text{num}}$; it is not a value of $\xi$ (R75). The HTML pages have been brought to the state after R148; in the process, calculation and programming errors in the Shor tools were also fixed, with no consequences for the classification. |
| R150 | Docs. 367, 353 (cf. R121, R148) | **Koide amplitude in Doc. 367; open question in Doc. 353 reformulated (9 October 2026).** Doc. 367 listed the amplitude factor $\sqrt{2}$ as proved, citing Doc. 353, and presented the Koide formula as completely parameter-free; Doc. 353 itself classifies this step as open (R148). Corrected: the phase $\theta = \lvert A\rvert^2$ remains the central result of Doc. 367; the amplitude is now treated there as a setting with the right value, the form $3\lvert A\rvert$ as not derived **[S]**, and the empirical exclusion of the exact pair from Doc. 369 is stated. In Doc. 353 the open question is reformulated: the Koide condition is equivalent to equal shares of the symmetric mode and the non-trivial pair, geometrically to the angle reading due to Foot **[B]**; what has to be justified is this equipartition **[S]**, no longer the choice between two assignments. The coupling weight $1/\sqrt{2}$ from Doc. 348 has a different origin; whether the two routes belong together is open. |

| No. | Affects | Subject (short form; details: Archive) |
|---|---|---|
| R151 | Docs. 360, 359 (cf. Doc. 358) | **HRV analysis in Doc. 360 recomputed; Theorems A and B in Doc. 359 refined (10 October 2026).** The Polar H10 analysis in Doc. 360 relied on a beat detection that missed a large share of beats in all paced-breathing recordings; joining the remaining intervals end to end had compressed the time axis and shifted all spectral frequencies. With validated detection and the time axis preserved, the dominant peak lies at the breathing frequency and the HF peak is its harmonic. The reported grid hits of LF/HF, the absent resonance effect and the absent coincidence of LF peak and respiration are withdrawn; grid proximity of LF/HF lies within the chance range of a null model. LF/HF is a power ratio and not a test object of the hypothesis, which concerns frequency ratios. In Doc. 359 the HRV band limits are compatible with the 5-limit grid but not determined by it (Theorem A); the resolution condition reads $\Delta f/f \leq 100\xi$, and an empirical counterpart in AI analyses is not established (Theorem B). Both documents are corrected directly. Open: prospective test of frequency coupling during paced breathing **[S]**. |
| R152 | Doc. 389 (cf. Docs. 312, 392) | **Equivalence of the three representations of gravitation delimited (11 October 2026).** Doc. 389 presented space curvature, change of mass or clock rate and fractal path lengthening as models that must be equivalent by virtue of the duality, and in its conclusion as equivalent models of the same situation, without restriction. The equivalence is shown for the static, non-rotating weak field with $\gamma = \beta = 1$ and for the conformally flat part of the geometry, to which the universe at large belongs. Around rotating masses the flat representation requires a co-moving vector field, for gravitational waves a tensor part; both are contained in the curvature representation through the adopted field equations (Doc. 312) and are not derived in the flat one (Doc. 392). Doc. 389 is refined directly in three places. Open: co-moving field and tensor part from the structure of the carrier **[S]**. |
| R153 | Docs. 312, 267, 308 (cf. Docs. 280, 392) | **Fourth cosmic discriminator computed; exponential form of the distance law excluded (11 October 2026).** Doc. 312 listed the forward demonstration of the four discriminators light curves, background temperature, Tolman and Alcock–Paczyński as a programme. The full mass reading passes the first three; the Alcock–Paczyński test does not separate expansion and the mass reading, because angular-diameter and luminosity distance follow in the mass reading as in the Friedmann model. It does, however, test the distance law: the pure exponential form of Docs. 280 and 308, read as coordinate distance of the mass reading, is excluded at high redshift by the ratio of transverse to radial distance, which is independent of the sound horizon **[X]**. The statement in Doc. 267 that the BAO data are degenerate between the two readings and neither confirm nor refute FFGFT therefore holds only for a distance law that departs from the exponential form at high redshift; no such law has been derived. Open: the distance law from the structure of the carrier **[S]**. |

| No. | Affects | Subject (short form; details: Archive) |
|---|---|---|
| R154 | Docs. 300, 384; narrative book ch. 23 (cf. R113, R121, R123, R135, R139, R140, R141) | **Uniformity of the fractal correction secured only as a principle (23 October 2026).** Docs. 300 and 384 present $K_{\text{frak}}$ as the single per-step factor of the scale recursion from Doc. 133, which appears only in the SI translation. In the actual calculations this uniformity is not achieved throughout: the factor demanded by $\alpha$ (R135, R139), the residual of the one-anchor chain for $G$ (R141), the lepton residuals (R113, R121) and the hadronic factor (R123) are separate residuals for each quantity that have not yet been traced back to the one recursion. In addition, ch. 23 of the narrative book calls the bridge factor from Doc. 158 $K_{\text{frak}}$ and describes it as universal; this contradicts R123 and is to be read in the sense of R140. The reading that the measurer stands inside the recursion, so that the residuals are part of the measured values, becomes robust only once a rule fixed in advance carries all SI translations at the same time. Open: a uniform recursion rule for all SI translations **[S]**. |
| R155 | Doc. 394 (cf. R112, R113, R121, R135, R139, R141, R154) | **R154 narrowed: the uniform rule can so far be tested only on $\alpha$ (23 October 2026).** Doc. 394 sorts the residuals named in R154 by origin. The residual for $G$ is an algebraic consequence of the residual of $E_P$ and, together with the $G$ formula, depends on fitted quantities (R141); the hadronic factor is a ratio of ladder coefficients, and $e^{-\xi}$ in Doc. 037 is a different quantity. The generation differences of the lepton residuals are the distance of the ladder from the Koide form and occur even without any measured value; they do not belong to the SI translation. For the ratios, the question from R113 is thereby answered in the sense of reading B (R121). The only independent test of the rule that remains is $\alpha$ (R135, R139). Open: a second test of the uniform rule on a quantity without a fitted factor, for instance $\tau_\mu$ (R112) **[S]**. |
| R156 | Docs. 395, 394 (cf. R112, R135, R154, R155) | **R155 closed: there is no second pilot in the SI sector (24 October 2026).** R155 listed a second test of the uniform rule on a quantity without a fitted factor as open and named the muon lifetime as a candidate. Doc. 395 shows that the only relation in the corpus with an SI unit is the bridge to $\alpha$; all other statements are ratios in which the factor cancels. The SI sector therefore has a single degree of freedom, in which $K$ and the MeV anchoring coincide, and $\alpha$ fixes it; a second test cannot exist. The muon lifetime tests the dimensionless ratio $v/m_\mu$ and hence the normalisation of the ladder (R112), not $K$. Doc. 394 has been made more precise directly at the relevant places. What remains open is whether the value of the factor is geometrically justified (R135) **[S]**. |
| R157 | Doc. 395 (cf. R142, R156) | **R142 made more precise: the gate deviation is in principle not measurable as long as it is common (24 October 2026).** R142 recorded that the systematic deviation of the $\pi$ gates is absorbed in the pulse calibration of real qubits and is therefore not directly measurable. Doc. 395 gives the reason: the calibration acts as the pilot of a channel estimation; a factor that is the same for all gates is a single degree of freedom and is absorbed completely, regardless of the quality of the hardware. Only a mode-dependent part would be visible; for it, Doc. 395 gives a bound from the lepton sector whose transfer to qubits is an analogy. Open: a mode-dependent gate factor from the structure of the carrier **[S]**. |

## .2 Currently open bridges (as of R157)

- $\tau_\mu$ directly from FFGFT, without detour via $G_F$ and $v$ — the only convention-free test of the weak sector (R112, Doc. 351) [S]
- Why do $\varepsilon_i$ decrease with generation? In Reading A (T0 fundamental) attributed to QED + SI projection, R113 bridge open **[S]**; in Reading B (Koide fundamental, Doc. 367) answered: distance cosine–power law, no QED needed **[K]**. What remains open in Reading B is why the rational approximation carries exactly the GF$(3^n)$ winding numbers (Doc. 369 § 4, R121) [S]
- Koide amplitude $d/c = \sqrt{2}$ from FFGFT geometry — orthogonal to $\theta = 2/9$ **[B]**, not yet anchored; equivalent to equal shares of the symmetric mode and the non-trivial pair, whose justification is open (Doc. 352, § 8; Docs. 353, 367; R148, R150) [S]
- Galois hypothesis for HRV frequency ratios: prospective coupling test during paced breathing (Doc. 360, R151) [S]
- Co-moving field and tensor part of gravitational waves in the flat representation from the structure of the carrier (Docs. 389, 392; R152) [S]
- Cosmic distance law from the structure of the carrier that passes the Alcock–Paczyński test (Docs. 267, 308, 392; R153) [S]
- Value of the SI factor: according to Doc. 395 the SI sector has only one degree of freedom, fixed by $\alpha$; open is whether $74/75$ is geometrically justified (Docs. 300, 384, 394, 395; R154–R156) [S]
- Mode-dependent gate factor from the structure of the carrier; a common factor is absorbed by the calibration (Doc. 395; R142, R157) [S]
- Charge quantisation $Q \in \{0, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1\}$: *closed by Doc. 346, R111 [B]* —
- Gell-Mann–Nishijima $Q = I_3 + Y/2$: charge quantisation closed (Doc. 346, R111 [B]); hypercharge of the charged leptons open (R146) [S] —
- $\Delta m^2_{32}$ atmospheric ($+16.0\,\%$): *closed by Doc. 340, R103 [K]* —
- Self-adjointness of $\hat{F}$ on the full space (beyond the $\mathbb{Z}_3$-invariant sector; R145, Doc. 327) [S]
- Geometric derivation of the hadronic factor in Doc. 160 ($K_{\text{had}}$, R123) [S]
- $\Lambda_{\text{QCD}}$ from $\xi$ — not derivable from the $\xi$ ladder (R122, Doc. 372) [X]; Higgs mass: candidates via Galois chain and trace rule (Doc. 373) [K]; residual of the trace rule presumably from the definition of $v$ (R112) [S]
- 2-loop corrections to $\alpha_s(M_Z)$ [S]
- $m_{\text{Pl}}$ and $\alpha_{\text{em}}(M_Z)$ from $\xi$ (external inputs in R81) [S]
- Quark/hadron sector — Doc. 318, R76
- Greybody factors; back-reaction on orbifold fixed points — R85 [S]
- Residual between the $K_{\text{frak}}$ required by $\alpha$ and $1 - 100\xi$; form of the correction factor against the candidate from Doc. 291 (R135, R139) [S]
- Formulate $T_{\text{CMB}}$ from $\xi$ independently of units and derive the coefficient in P11 (R137, R70)
- Physical Casimir signature of $L_\xi$ (R136)
- Structural explanation of the residual of the one-anchor chain for $G$ (R141, Doc. 383) [S]
- Forward derivation of the cosmic exponent $41/4$ (P20); the length in $1 + z = e^{\xi x}$ also hangs on it (R137); the $10^{123}$ question, $R_H \to \ell_1$ and $\kappa = \Lambda_{\text{obs}} R_H^2$ hang on it

- Light elements without a hot early phase: abundances of $^4$He and D as dimensionless number ratios, not reinterpretable via frequency degeneracy (Doc. 267, 222; R128) [S]
- CMB peak selection: generate $\{1, 6, 14, 26\}$ forward; the $|n|^2{=}30$ case (P29/P31); the factor 3 is a setting, not a projection (R139)
- $C_\ell$ projection: physical source/window function (P30)
- $\Omega_{\mathrm{DM}}$ quantitatively, model-neutral (P17)
- Spectral dimension $d_s = 1.86$ vs. FFGFT $\approx 2$
- Corpus-wide $H_0$ language cleanup (P34); fundamental rework of Doc. 026 (P16)
- Justify the base mass $M_{\mathrm{base}} = m_P$ in the fundamental sector — Doc. 180, R90 [S]
- $\xi$ as line width (P44, pre-registered); $\iota$ embedding HLV $\subset$ FFGFT in the time sector (Doc. 297)
- Adversarial $D_4$ operator-specificity test against matched comparison carriers (GAE logic) — R95 [S]

**R86 – Self-adjointness of $\hat{F}$ (August 2026):** Doc. 327 closes the gap declared as [S] in R82 on the $\mathbb{Z}_3$-invariant sector $P_0$ (refined by R145). The finiteness of the scale ladder is guaranteed by $L_0 = \xi \cdot \ell_P$ (Doc. 180 **[K]**) – no infinities in FFGFT, hence $\hat{F}$ is bounded and $\mathscr{D}(\hat{F}) = \mathscr{H}$. The $\mathbb{Z}_3$ pairing $(k, -k)$ (the same as in Doc. 325 **[B]**) gives $\varphi_{-k} = \bar{\varphi}_k$, hence normality; $\hat{F}$ is symmetric on $P_0$. Deficiency indices $(0, 0)$ on $P_0$: exactly one self-adjoint realisation on $P_0$ **[B]**. Fractal measure (finite 100-step recursion), $\mathbb{Z}_3$ restriction and all three $\chi$-twist classes preserve self-adjointness on $P_0$ **[B]**. $\xi = \lambda_{\mathrm{min}}(\hat{F}_{D_4})$ well-defined and truncation-stable (min-max), provided the $D_4$ sector lies in $P_0$ ($D_4 \subset P_0$) **[B]**. On the full space only $(\hat{F} + \hat{F}^\dagger)/2$ is self-adjoint; using it is a choice (R145). Verification script: `327_selbstadjungiertheit_F_verify.py` (26 assertions).
*Declared 23 August 2026*

**R87 – $\Delta m^2_{32}$: corrected numerical values and candidate analysis (August 2026):** In Doc. 320 the evaluation of $\xi^{9/5}$ was faulty ($8.717{\times}10^{-8}$ instead of $1.059{\times}10^{-7}$); the mass formula $m_{\nu_i} = m_e\, \xi^{p_i}$ and the exponents $p_i \in \{9/4,\ 2,\ 9/5\}$ were always correct. Docs. 320 and 322 have been brought to the correct numerical values: $m_{\nu_3} = 54.11$ meV **[K]**, $\Delta m^2_{32} = 2.846 \times 10^{-3}$ eV² **[K]** (deviation $+16.0\,\%$ against NuFIT 5.3), $\sum m_\nu = 64.17$ meV **[K]** (still within the Planck limit). Unaffected: $m_{\nu_1}$, $m_{\nu_2}$, $\Delta m^2_{21}$ ($+8.3\,\%$) and the fixed-point assignment. Doc. 320 was at the same time extended by a candidate analysis for the remaining deviation: $K_{\mathrm{frak}}$ at fixed point $F_7$ ($+12.8\,\%$, insufficient), mixing term $F_5$–$F_7$ (right sign, complete mass matrix outstanding – the most promising candidate), alternative exponents (unlikely: $p_3^{\mathrm{PDG}} = 1.8081$ lies only $0.45\,\%$ above $9/5$). $\Delta m^2_{32}$ remains **[S]** here; closed by Doc. 340 (R103). Verification script: `320_dm32_neutrino_verify.py` (10 assertions).
*Declared 23 August 2026*

**R88 – $n_{\mathrm{thresh}}$ is scale-dependent; Matzke's scale reconstructed (August 2026):** Doc. 329 settles numerically the question booked as [S] in Doc. 326. The collapse condition Compton = Schwarzschild yields $M_{\mathrm{coll}} = m_P/\sqrt{2}$ purely geometrically and free of any bit value **[B]**. With the system-dependent bit energy $E_{\mathrm{bit}}(L) = \hbar c/L$ (Doc. 257) one obtains exactly $n_{\mathrm{thresh}}(L) = L/(\sqrt{2}\,\ell_P)$ – linear in the scale **[B]**. Matzke's value $6.4097$ corresponds exactly to $L = 2\pi\ell_P/\ln 2 = 9.0647\,\ell_P$ **[B]**; the "universal" bit value is the FFGFT bit energy at that scale. Across the scales of the corpus $n_{\mathrm{thresh}}$ varies by a factor of $4.6 \times 10^{29}$; a $10\,\%$ temperature deviation flips the stability claim for $\mathrm{Cl}(6)$. Hence $n_{\mathrm{thresh}} = 6.41$ as a universal number is negatively closed **[X]**. Untouched: Compton = Schwarzschild, $r_s$ without field equations, $\mathrm{Cl}(6)$ structure **[B]**. Verification script: `326_nthresh_skalenabhaengigkeit.py` (12 assertions).
*Declared 24 August 2026*

**R89 – Time-field mediator mass: one structure, sector-dependent base mass (August 2026):** The corpus uses $m_T$ with widely differing numerical values (Docs. 019/180/201: $m_T = \lambda/\xi$; Doc. 032: effective torsion mass $\approx 5.220$ GeV). This is no contradiction: both follow the rule $m_T = M_{\text{base}}/\xi$, whence the range of the massive time field is $\hbar/(m_T c) = \xi \cdot \lambda_C(M_{\text{base}})$ **[B]** — universal form, sector-dependent value, structurally identical with $E_{\text{bit}} = \hbar c/L$ (Docs. 257, 329). Lepton sector ($M = m_e$): $m_e/\xi = 3.83$ GeV, with the $O(1)$ factors of Doc. 032 ($\sin \pi\xi$, $\pi^2$, $\sqrt{\alpha/K_{\text{frak}}}$, $R_f$) 5.220 GeV — range $3.78\times10^{-17}$ m. *(The 5.220 GeV follow only with the $E_0 = 7500$ GeV set in Doc. 032; cf. the notes in Doc. 032 of 30 Sept. 2026 and Doc. 329 of 1 Oct. 2026.)* Fundamental sector ($M = m_P$): $m_T = m_P/\xi = 7500\, m_P$ — range $2.155 \times 10^{-39}$ m $= L_0$ exactly. Doc. 032 itself calls its quantity an *effective* torsion mass; sameness of name is not sameness of quantity. Verification script: `180_mT_struktur_L0_verify.py` (17 assertions).
*Declared 24 August 2026*

*Placement after Doc. 287:* The Lagrangian description ($m_T = M_{\text{base}}/\xi$) and the bit-energy description ($E_{\text{bit}} = \hbar c/L$) are *different representations of the same floor structure* – in the terms of Doc. 287, presumably a **bijection**: lossless and mutually translatable, but neither more complete than the other. The generative core that loses nothing is the $T^4$ torus (Doc. 287, Doc. 243); all other descriptions stand to it in one of three relations: identity, bijection or projection.

**R90 – The exponent in $L_0 = \xi \cdot \ell_P$ is derived (August 2026):** Doc. 180 justifies $L_0 = \xi \cdot \ell_P$ by the $\xi$ scaling of the coupling term $g_T^{\ell} = \xi m_{\ell}$ — an argument by analogy. The exponent however follows *necessarily* from the Lagrangian mass term: $m_T = M_{\text{base}}/\xi^1$ together with $\lambda_C \sim 1/m$ gives range $\sim \xi^1$ **[B]** (R89). Counter-check: $m_T \sim \xi^{-n}$ would give $L_0 = \xi^n \ell_P$; the exponent of the range is the negative exponent of $m_T$. $L_0$ is thus the Compton range of the time-field mediator in the fundamental sector — below it the time field can no longer mediate, which is why "distance" loses its meaning. This also explains *why* a floor exists at all: because the mediator is massive. What remains open is only the choice $M_{\text{base}} = m_P$ for the fundamental sector; it introduces no new parameter ($\lambda = 1$ in Planck units) but is not independently justified [S]. Without consequences for Doc. 327 (R86): boundedness requires only a finite floor **[B]**.
*Declared 24 August 2026*

**R91 – $\lambda$ in $m_T = \lambda/\xi$ is a base mass (August 2026):** Docs. 019 and 201 describe $\lambda$ as the "Higgs coupling parameter". This is misleading: (i) Dimensionally $\lambda$ must be a *mass*, since $m_T$ is a mass and $\xi$ is dimensionless **[B]**; the Higgs quartic $\lambda_h \approx 0.129$ is dimensionless. (ii) $\lambda_h$ runs under the RGE and is a low-energy parameter of the electroweak scale; a value measured there cannot be inserted at the Planck scale. (iii) Interpreted as a dimensionless factor in Planck units, $\lambda_h$ would give the range $7.75 \cdot L_0$ — missing $L_0$. Correct is $\lambda = M_{\text{base}}$ of the respective sector, in the fundamental case $m_P$ (R89/R90). A Higgs connection can at most be an effective statement in the electroweak sector, never the definition. The label in Docs. 019/201 is to be corrected [S] *(done: addendum R91 in Docs. 019 and 201, 7 Sept. 2026; Doc. 019 corrected in the text on 30 Sept. 2026)*.
*Declared 24 August 2026*

**R92 – Two misattributions concerning the time field (August 2026):** (i) *Doc. 250*: The sentence "The term $-\frac{1}{2}m_T^2(\partial m)^2$ with $m_T = \xi/\ell$ (Doc. 201)" contains two errors. In natural units $m = 1/\ell$, so $\xi/\ell = \xi \cdot m$ — this is the *coupling* $g_T^{\ell} = \xi m_{\ell}$, not the mediator mass $m_T = M_{\text{base}}/\xi$ **[B]**. Furthermore the mass term reads $-\frac{1}{2}m_T^2(\Delta m)^2$, not $(\partial m)^2$; the latter would contribute to the kinetic term. The physical point (the mass term provides the stiffness against divergence of $\Delta m$) remains correct; only the formula is misquoted [S]. (ii) *Doc. 190, Clarification 4*: The statement recorded as an internal consistency check, that an object with $M_0 = \xi m_P/2$ would have Schwarzschild radius exactly $L_0$, is an *identity*: from $r_s = 2GM/c^2$ and $\ell_P = G m_P/c^2$ it follows that $r_s(x m_P/2) = x\ell_P$ for any $x$ and any exponent **[B]**. It tests neither $\xi$ nor the exponent

in $L_0$; classifying it as a "check" overstates it. Non-trivial would be only the condition appended there, whether $M_0$ coincides with a rung of the $\xi$ hierarchy — open [S].
*Declared 24 August 2026*

**R93 – $S_{\mathrm{BH}}(M_{\mathrm{coll}}) = 2\pi$ nat $= n_{\mathrm{thresh}}$ (Matzke) in bits (August 2026; refined by R144: $= L_{\mathrm{Matzke}}/\ell_P$, $n_{\mathrm{thresh}} = S_{\mathrm{BH}}/\sqrt{2}$):** Doc. 325 §5.3 (new) and Doc. 329 **[B]** jointly yield: $r_s(M_{\mathrm{coll}}) = \sqrt{2}\,\ell_P$ (geometric, Compton = Schwarzschild); $S_{\mathrm{BH}}(M_{\mathrm{coll}}) = 4\pi \cdot 2\ell_P^2/(4\ell_P^2) = 2\pi$ nat exactly. In bits: $2\pi/\ln 2 = 9.0647$ bit $= n_{\mathrm{thresh}}$ (Matzke) **[B]** (refined by R144: $= L_{\mathrm{Matzke}}/\ell_P$; $n_{\mathrm{thresh}} = S_{\mathrm{BH}}/\sqrt{2} = 6.41$). Interpretation: Matzke's stability threshold is the residual entropy of the evaporation endpoint — the same geometry in two languages. Matzke's universality claim remains closed negative **[X]** (R88); this identity explains *why* the value $9.065$ appears in both contexts without making it universal. Verification script: `325_informationsbilanz.py` (9 assertions, assertions 2–3).
*Declared 25 August 2026*

**R94 – $I_{\mathrm{sector}}/I_{\mathrm{thermal}} = \log_2 3$ for all $M$ (August 2026; refined by R144: $\ln 3$):** From Doc. 325 §5.1 **[B]** ($\log_2 3$ bits of sector information per quantum, orthogonal $\mathbb{Z}_3$ pair states) and §Building Block 2 **[B]** (1 nat thermal area bookkeeping per quantum via Clausius):

$$\frac{I_{\mathrm{sector}}}{I_{\mathrm{thermal}}} = \log_2 3 \approx 1.585 \quad \text{for every mass } M > M_{\mathrm{coll}}.$$

The ratio is mass-independent — a structural constant of the $T^4/\mathbb{Z}_3$ geometry. Total budget: $N_{\mathrm{emit}} = S_{\mathrm{BH}}(M_{\mathrm{init}}) - 2\pi$ nat, $I_{\mathrm{sector}} = N_{\mathrm{emit}} \cdot \log_2 3$ bits. Verification script: `325_informationsbilanz.py` (assertions 6–7).
*Declared 25 August 2026*

**K-041-a/b/c – Internal inconsistencies in Doc. 041 (August 2026):** Doc. 041 remains unchanged (Doc. 041 has carried corrections and notes in the text since 30 Sept. 2026, R133/R135). The following notes document places where printed formulas and table values in the same document disagree under the document's own $\xi = (4/3) \cdot 10^{-4}$ (Doc. 328 §7.4b). Resolution requires author review of the source document.

**K-041-a** ($\delta_{\mathrm{CKM}}$): Formula $\delta_{\mathrm{CKM}} = \arcsin\left(2\sqrt{2} \cdot \sqrt{\xi}/3\right)$ yields $0.0109$ rad; table value $1.20$ rad. Candidate: $\arcsin\left(2\sqrt{2}/3\right) = 1.231$ rad — factor $\sqrt{\xi}$ likely a typesetting error.

**K-041-b** ($\theta_{13}$): Formula $\theta_{13} = \arcsin\left(\xi^{1/3}\right)$ yields $2.93°$; table value $8.57°$. Candidate: argument $1/300 = 25\xi$ yields $8.59°$ — whether $1/300$ is a corpus quantity is to be clarified at the source document (clarified: $25\xi = 1/300 = 1/(5|A_5|)$ is the corpus quantity from Docs. 328/340, R103).

**K-041-c** ($|V_{us}|$): Formula with Doc. 006 masses yields $0.2124$; table value $0.22452$ (dev. $5.4\,\%$). Possible cause: differing masses or $f_{\mathrm{Cab}}$ definition in Doc. 041. *(K-041 passages in Doc. 041 corrected on 1 Oct. 2026: $\delta_{\mathrm{CKM}}$ $1.2 \to 0.0109$ rad; $\theta_{13}$ $8.6° \to 2.93°$, $8.59°$ only with $(25\xi)^{1/3}$; $|V_{us}|$ $0.225 \to 0.21$; notes of 1 Oct. 2026 in Doc. 041.)*
*Declared 25 August 2026*

**R95 – Doc. 330: two wording refinements and Type-I clarification after external review (August 2026):** External review (M. Krüger, IPI, 25 August 2026) of Doc. 330 identified two overly superficial formulations; the corpus itself (Docs. 270/314/327, R41, R86) was correct and more precise in each case. Doc. 330 remains unchanged; the refinements apply at the register level (Doc. 330 has carried notes since 1 Oct. 2026, R145).

*(i) Intrinsic flatness of $T^4$ (Doc. 330 §2.2).* **Too superficial:** "positive and negative curvature contributions cancel in the sum" — this conflates the *embedded* torus in $\mathbb{R}^3$ ($K > 0$ outside, $K < 0$ inside, only globally $\int K\,dA = 0$ via Gauss–Bonnet, $\chi = 0$) with the *flat* torus $\mathbb{R}^4/\mathbb{Z}^4$. **Precise:** The flat metric $\delta_{ij}$ is $\mathbb{Z}^4$-invariant and descends undistorted to the quotient; $\Gamma^k_{ij} = 0$, Riemann tensor pointwise zero, $K = 0$ everywhere — there is nothing to cancel. Triply

verified constructively ($\Gamma = 0$, angle sum $\pi$, trivial holonomy). The conclusion of Doc. 330 (intrinsically flat, Fourier basis without curvature corrections, Type III bijective) remains unchanged **[B]**. Nash–Kuiper side note: the flat torus is not $C^2$-isometrically embeddable in $\mathbb{R}^3$ — the embedding question does not arise for the fundamental level anyway (Nash–Kuiper remark refined by R96).

*(ii) Self-adjointness of $\hat{F}$ (Doc. 330 §Question 3).* **Too superficial:** "bounded and therefore self-adjoint" — boundedness alone does not imply self-adjointness (counterexample: shift operator). **Precise (= R86 [B]):** bounded *plus* symmetric via the $\mathbb{Z}_3$ pairing $(k, -k)$ on a complete Fourier basis ⇒ self-adjoint; deficiency indices $(0, 0)$; boundedness ($\|\hat{F}\| \le \sum_n \xi^n = \xi/(1-\xi)$, convergent even for $N \to \infty$) secures $\mathscr{D}(\hat{F}) = \mathscr{H}$, while the symmetry argument carries the burden independently. Status **[B]** unchanged. *(Refined by R145: self-adjoint only on the $\mathbb{Z}_3$-invariant sector; $r_n = \xi^n$ is not a corpus choice, cf. the notes in Doc. 330 of 1 Oct. 2026.)*

*(iii) Type-I direction clarification (no error; confirmation of R41).* The review objection that the unrolling $S^1_m \to \mathbb{R}_t$ has a lifting problem (covering $p : \mathbb{R} \to S^1$ without canonical inverse) concerns the *lifting of points* — Type I, however, lifts no points but is the *function pullback* $\Phi = p^* : L^2(S^1_m) \to \mathrm{Per}_{R_m}(\mathbb{R}_t)$, $(\Phi f)(t) = f(t \bmod R_m)$ — canonical, no additional datum. Precisely the reverse direction $\mathbb{R} \to S^1$ is the lossy one (R41, declared 19 June 2026). The external six-point checklist is answered completely: injective; surjective onto $\mathrm{Per}_{R_m}$ (not onto $L^2(\mathbb{R}_t)$ — that *is* the embedding statement of Doc. 270); isometry per period (Haar ↔ Lebesgue/period); inverse = restriction to an arbitrary period (period choice = deck-transformation gauge freedom, not a missing datum); spectrum discrete ↔ discrete frequencies $n/R_m$; $\mathrm{U}(1)$ translation ↔ time translation commute with $\Phi$. The proof sketch of Doc. 270 (theorem "Type I is information-preserving in mode content") is thereby completed to a full theorem: status change [sketch] → **[B]**.

*(iv) Accepted as open:* An adversarial operator-specificity test for $D_4$ against matched comparison carriers (GAE logic) does not exist; $5 \nmid |\mathrm{Aut}(D_4)| = 1152$ is a lattice-specific algebraic fact (excluding e.g. $E_8$ with $5 \mid |\mathrm{Aut}(E_8)|$), not a specificity proof. Carried as an open bridge **[S]**.

Verification scripts: `pruef_krueger_einwaende.py` (16/16), `pruef_k1_tief.py` (8/8), `pruef_k3_tief.py` (6/6), `pruef_k4_tief.py` (5/5).
*Declared 25 August 2026*

**R96 – Type-I follow-up refinement: pullback is a change of representation, not a new sector (August 2026):** Second feedback (M. Krüger, IPI, 26 August 2026) on the R95(iii) clarification accepts the function pullback $\Phi = p^*$ as the correct mathematical object and the refutation of the original lifting objection, but refines a remaining point: $\Phi f$ is $R_m$-periodic, so $\int_{\mathbb{R}} |\Phi f|^2 \, dt = \infty$ for any $f \neq 0$ — $\Phi f$ does not lie in ordinary $L^2(\mathbb{R}_t)$ with Lebesgue measure over the whole line. The correct target space $\mathrm{Per}_{R_m}(\mathbb{R}_t)$ (periodic $L^2_{\mathrm{loc}}$, norm over one period) is isometrically isomorphic to $L^2(S^1_m)$ itself — the same countable Fourier basis $\{e_n\}_{n \in \mathbb{Z}}$, the same Parseval identity. Formally:

$$L^2(S^1_m) \;\cong\; \mathrm{Per}_{R_m}(\mathbb{R}_t) \quad \text{(per-period norm)}, \qquad \textit{not} \qquad L^2(S^1_m) \;\cong\; L^2(\mathbb{R}_t).$$

Algebraically confirmed (`pruef_332_krueger_periodizitaet.py`): divergence of the whole-line integral for any nontrivial periodic function (translation-invariant Lebesgue measure, each period contributes the same positive amount); isomorphism of the Fourier bases of both spaces via $\langle e_{n_1}, e_{n_2} \rangle = R_m \, \delta_{n_1 n_2}$ over one period. Status **[B]**.

**Assessment:** This is not a correction to Doc. 330/R95(iii), but a precisification of what the status change [sketch]→[B] there mathematically carries: an information-preserving representation of the compact sector on the universal cover, not a genuinely noncompact state space. Doc. 332 (Section 2, "canonical function pullback") already formulated this

restraint correctly (“change of representation of the same function algebra”, no extension), but without explicitly separating the norm/Hilbert-space level from the algebra level; Doc. 332 was accordingly extended with a clarifying paragraph (Section 2.1, “continuum of the coordinate representation” vs. “continuum of the state space”) and recompiled.

*Side correction (Nash–Kuiper):* The Nash–Kuiper side note in Doc. 330/R95(i) is refined: Nash–Kuiper is an existence theorem for $C^1$-isometric embeddings, not a non-existence theorem for $C^2$ embeddings of the flat 2-torus in $\mathbb{R}^3$ — these two statements (existence of $C^1$ isometries and the $C^2$ obstruction question) must be kept separate. For the fundamental level $T^4$ the embedding question into $\mathbb{R}^3$ does not arise dimensionally anyway; the point was not load-bearing for the main R95(i) conclusion (intrinsic flatness) and does not change its status **[B]**.

Verification script: `pruef_332_krueger_periodizitaet.py` (4/4 assertions).
*Declared 26 August 2026*

*Declared 25 August 2026*